Code

Spatial Transcriptomics - Seurat Analysis of Human Glioblastoma (GBM) FFPE Tissue

Setting up environment for Spatial Analysis

Packages/Dependencies

Hide

```
library(Seurat)
library(SeuratData)
library(ggplot2)
library(patchwork)
library(dplyr)
library(Rfast2)
```

Loading 10x Visium Dataset

Hide

```
data_dir <- "C:/Users/hugor/OneDrive/Desktop/Bioinformatics/Spatial Transcriptomics/Glioblast
oma Spatial Data"

list.files(data_dir)
```

```
[1] "CytAssist_11mm_FFPE_Human_Glioblastoma_metrics_summary.csv" "filtered_feature_bc_matrix.
h5"
[3] "spatial"
```

Hide

```
install.packages("hdf5r")
```

```
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/hdf5r_1.3.16.zip'
Content type 'application/zip' length 4010902 bytes (3.8 MB)
downloaded 3.8 MB
```

```
package ‘hdf5r’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in
    C:\Users\hugor\AppData\Local\Temp\RtmpmUEwCn\downloaded_packages
```

Hide

```
library(hdf5r)

GBM <- Load10X_Spatial(data.dir = data_dir)

GBM
```

```
An object of class Seurat
18085 features across 10878 samples within 1 assay
Active assay: Spatial (18085 features, 0 variable features)
 1 layer present: counts
 1 spatial field of view present: slice1
```

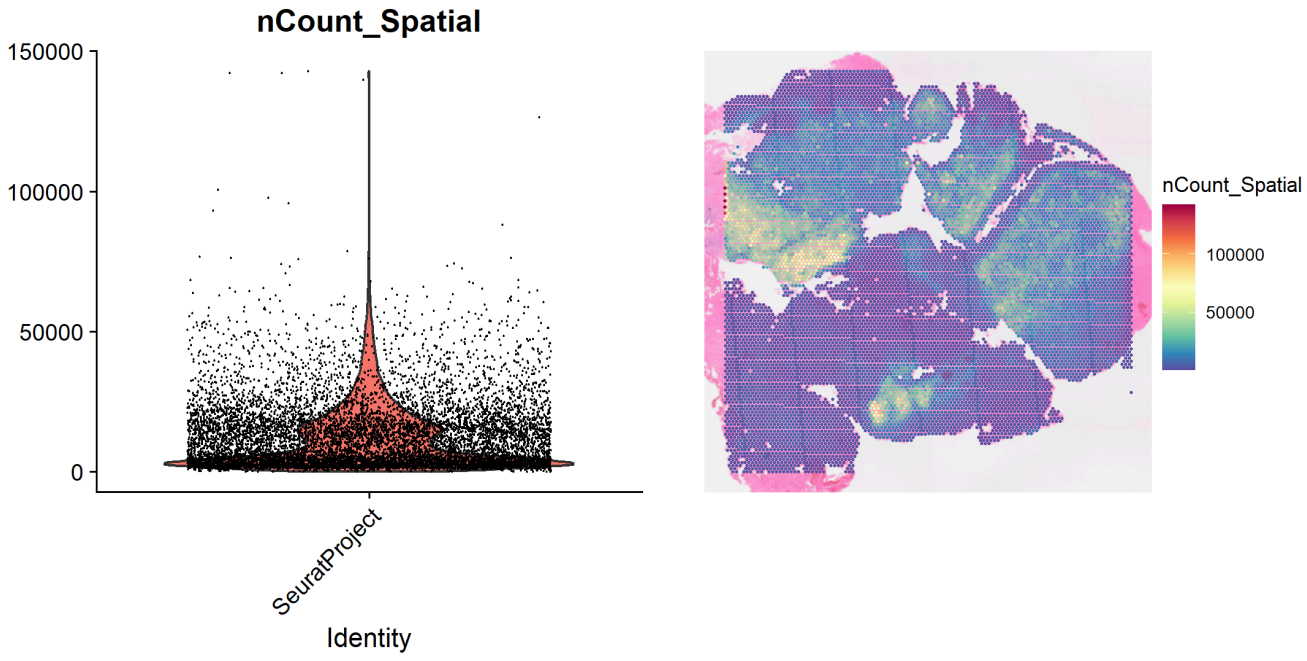
Data preprocessing

Hide

```
plot1 <- VlnPlot(GBM, features = "nCount_Spatial", pt.size = 0.1) + NoLegend()

plot2 <- SpatialFeaturePlot(GBM, features = "nCount_Spatial", pt.size.factor = 2.5) + theme(l
egend.position = "right")

wrap_plots(plot1, plot2)
```



Spatial Normalisation

Hide

```
GBM <- SCTransform(GBM, assay = "Spatial", verbose = FALSE)
```

Gene expression visualisation

Hide

```
features <- rownames(GBM)

features
```

```
  [1] "SAMD11"     "NOC2L"      "KLHL17"     "PLEKHN1"    "PERM1"      "HES4"       "ISG15"
"AGRN"       "RNF223"     "C1orf159"   "TTLL10"
 [12] "TNFRSF18"   "TNFRSF4"    "SDF4"       "B3GALT6"    "C1QTNF12"   "UBE2J2"     "SCNN1D"
"ACAP3"      "PUSL1"      "INTS11"     "CPTP"
 [23] "TAS1R3"     "DVL1"       "MXRA8"      "AURKAIP1"   "CCNL2"      "ANKRD65"    "TMEM88
B"    "VWA1"       "ATAD3C"     "ATAD3B"     "ATAD3A"
 [34] "TMEM240"    "SSU72"      "FNDC10"     "MIB2"       "MMP23B"     "CDK11B"     "CDK11A"
"NADK"       "GNB1"       "CALML6"     "TMEM52"
 [45] "CFAP74"     "GABRD"      "PRKCZ"      "FAAP20"     "SKI"        "RER1"       "PEX10"
"PLCH2"      "PANK4"      "HES5"       "TNFRSF14"
 [56] "PRXL2B"     "MMEL1"      "ACTRT2"     "PRDM16"     "ARHGEF16"   "MEGF6"      "TPRG1L"
"WRAP73"     "TP73"       "CCDC27"     "SMIM1"
 [67] "LRRC47"     "CEP104"     "DFFB"       "C1orf174"   "AJAP1"      "NPHP4"      "KCNAB2"
"CHD5"       "RNF207"     "ICMT"       "GPR153"
 [78] "ACOT7"      "HES2"       "ESPN"       "TNFRSF25"   "PLEKHG5"    "NOL9"       "TAS1R1"
"ZBTB48"     "KLHL21"     "PHF13"      "THAP3"
 [89] "DNAJC11"    "CAMTA1"     "VAMP3"      "PER3"       "UTS2"       "TNFRSF9"    "ERRFI1"
"SLC45A1"    "RERE"       "ENO1"       "CA6"
[100] "SLC2A7"     "SLC2A5"     "GPR157"     "H6PD"       "SPSB1"      "SLC25A33"   "TMEM20
1"    "PIK3CD"     "CLSTN1"     "CTNNBIP1"   "LZIC"
[111] "NMNAT1"     "RBP7"       "UBE4B"      "KIF1B"      "PGD"        "CENPS"      "CORT"
"DFFA"       "PEX14"      "CASZ1"      "C1orf127"
[122] "MASP2"      "SRM"        "EXOSC10"    "MTOR"       "ANGPTL7"    "UBIAD1"     "DISP3"
"FBXO2"      "FBXO44"     "FBXO6"      "MAD2L2"
[133] "DRAXIN"     "AGTRAP"     "C1orf167"   "MTHFR"      "CLCN6"      "NPPA"       "NPPB"
"KIAA2013"   "PLOD1"      "MFN2"       "MIIP"
[144] "TNFRSF8"    "TNFRSF1B"   "VPS13D"     "DHRS3"      "AADACL4"    "AADACL3"    "C1orf15
8"   "PRAMEF12"   "PRAMEF1"    "PRAMEF11"   "PRAMEF2"
[155] "PRAMEF4"    "PRAMEF10"   "PRAMEF6"    "PRAMEF27"   "HNRNPCL2"   "PRAMEF13"   "PRAMEF1
8"   "PRAMEF8"    "PRAMEF33"   "PRAMEF15"   "PRAMEF14"
[166] "PRAMEF19"   "PRAMEF20"   "LRRC38"     "PDPN"       "PRDM2"      "KAZN"       "TMEM51"
"FHAD1"      "EFHD2"      "CTRC"       "CELA2A"
[177] "CELA2B"     "CASP9"      "DNAJC16"    "AGMAT"      "DDI2"       "PLEKHM2"    "SLC25A3
4"   "TMEM82"     "FBLIM1"     "UQCRHL"     "SPEN"
[188] "ZBTB17"     "SRARP"      "HSPB7"      "CLCNKA"     "CLCNKB"     "FAM131C"    "EPHA2"
"ARHGEF19"   "CPLANE2"    "FBXO42"     "SZRD1"
[199] "SPATA21"    "NECAP2"     "NBPF1"      "CROCC"      "MFAP2"      "ATP13A2"    "SDHB"
"PADI2"      "PADI1"      "PADI3"      "PADI4"
[210] "PADI6"      "RCC2"       "ARHGEF10L"  "ACTL8"      "IGSF21"     "KLHDC7A"    "PAX7"
"TAS1R2"     "ALDH4A1"    "IFFO2"      "UBR4"
[221] "EMC1"       "MRTO4"      "AKR7A2"     "SLC66A1"    "CAPZB"      "MICOS10"    "NBL1"
"HTR6"       "TMCO4"      "RNF186"     "OTUD3"
[232] "PLA2G2E"    "PLA2G2A"    "PLA2G5"     "PLA2G2D"    "PLA2G2F"    "PLA2G2C"    "UBXN10"
"VWA5B1"     "CAMK2N1"    "MUL1"       "FAM43B"
[243] "CDA"        "PINK1"      "DDOST"      "KIF17"      "SH2D5"      "HP1BP3"     "EIF4G3"
"ECE1"       "NBPF3"      "ALPL"       "RAP1GAP"
[254] "USP48"      "HSPG2"      "CELA3B"     "CELA3A"     "WNT4"       "ZBTB40"     "EPHA8"
"C1QA"       "C1QC"       "C1QB"       "EPHB2"
[265] "LACTBL1"    "TEX46"      "KDM1A"      "LUZP1"      "HTR1D"      "HNRNPR"     "ZNF436"
"TCEA3"      "ASAP3"      "E2F2"       "ID3"
[276] "ELOA"       "PITHD1"     "GALE"       "HMGCL"      "FUCA1"      "CNR2"       "PNRC2"
"SRSF10"     "MYOM3"      "IL22RA1"    "IFNLR1"
[287] "GRHL3"      "STPG1"      "NIPAL3"     "RCAN3"      "NCMAP"      "SRRM1"      "CLIC4"
"RUNX3"      "SYF2"       "RSRP1"      "RHD"
[298] "TMEM50A"    "RHCE"       "MACO1"      "LDLRAP1"    "MAN1C1"     "SELENON"    "MTFR1L"
```

```
"AUNIP"      "PAQR7"      "STMN1"      "PAFAH2"
 [309] "EXTL1"      "SLC30A2"    "TRIM63"     "PDIK1L"     "FAM110D"    "C1orf232"   "AL39165
0.1" "ZNF593"     "CNKSR1"     "CATSPER4"   "CEP85"
 [320] "SH3BGRL3"   "UBXN11"     "CD52"       "CRYBG2"     "ZNF683"     "LIN28A"     "DHDDS"
"ARID1A"     "PIGV"       "ZDHHC18"    "SFN"
 [331] "GPN2"       "GPATCH3"    "NUDC"       "NR0B2"      "KDF1"       "TRNP1"      "TENT5B"
"SLC9A1"     "WDTC1"      "TMEM222"    "SYTL1"
 [342] "MAP3K6"     "FCN3"       "CD164L2"    "GPR3"       "WASF2"      "AHDC1"      "FGR"
"IFI6"       "FAM76A"     "STX12"      "PPP1R8"
 [353] "THEMIS2"    "RPA2"       "SMPDL3B"    "XKR8"       "EYA3"       "PTAFR"      "DNAJC8"
"ATP5IF1"    "SESN2"      "MED18"      "PHACTR4"
 [364] "RCC1"       "TRNAU1AP"   "TAF12"      "RAB42"      "GMEB1"      "YTHDF2"     "OPRD1"
"EPB41"      "TMEM200B"   "SRSF4"      "MECR"
 [375] "PTPRU"      "MATN1"      "LAPTM5"     "SDC3"       "PUM1"       "NKAIN1"     "SNRNP4
0"    "ZCCHC17"    "FABP3"      "SERINC2"    "TINAGL1"
 [386] "HCRTR1"     "PEF1"       "COL16A1"    "ADGRB2"     "SPOCD1"     "KHDRBS1"    "TMEM39
B"    "KPNA6"      "TXLNA"      "CCDC28B"    "IQCC"
 [397] "DCDC2B"     "TMEM234"    "EIF3I"      "FAM167B"    "LCK"        "HDAC1"      "MARCKSL
1"   "TSSK3"      "FAM229A"    "BSDC1"      "ZBTB8B"
 [408] "ZBTB8A"     "RBBP4"      "SYNC"       "KIAA1522"   "YARS"       "S100PBP"    "FNDC5"
"HPCA"       "TMEM54"     "RNF19B"     "AZIN2"
 [419] "TRIM62"     "ZNF362"     "A3GALT2"    "PHC2"       "ZSCAN20"    "CSMD2"      "HMGB4"
"C1orf94"    "SMIM12"     "GJB5"       "GJB4"
 [430] "GJB3"       "GJA4"       "DLGAP3"     "TMEM35B"    "ZMYM6"      "ZMYM1"      "SFPQ"
"ZMYM4"      "KIAA0319L"  "NCDN"       "TFAP2E"
 [441] "PSMB2"      "C1orf216"   "CLSPN"      "AGO4"       "AGO1"       "AGO3"       "TEKT2"
"ADPRHL2"    "COL8A2"     "TRAPPC3"    "MAP7D1"
 [452] "THRAP3"     "SH3D21"     "EVA1B"      "STK40"      "LSM10"      "OSCP1"      "CSF3R"
"GRIK3"      "ZC3H12A"    "MEAF6"      "SNIP1"
 [463] "DNALI1"     "GNL2"       "RSPO1"      "C1orf109"   "CDCA8"      "EPHA10"     "MANEAL"
"YRDC"       "C1orf122"   "MTF1"       "INPP5B"
 [474] "SF3A3"      "FHL3"       "UTP11"      "POU3F1"     "RRAGC"      "MYCBP"      "GJA9"
"RHBDL2"     "AKIRIN1"    "NDUFS5"     "MACF1"
 [485] "BMP8A"      "PABPC4"     "HEYL"       "NT5C1A"     "HPCAL4"     "PPIE"       "BMP8B"
"OXCT2"      "TRIT1"      "MYCL"       "MFSD2A"
 [496] "CAP1"       "PPT1"       "RLF"        "TMCO2"      "ZMPSTE24"   "COL9A2"     "SMAP2"
"ZFP69B"     "ZFP69"      "EXO5"       "ZNF684"
 [507] "RIMS3"      "NFYC"       "KCNQ4"      "CITED4"     "CTPS1"      "SLFNL1"     "SCMH1"
"FOXO6"      "EDN2"       "HIVEP3"     "GUCA2B"
 [518] "GUCA2A"     "FOXJ3"      "RIMKLA"     "ZMYND12"    "PPCS"       "CCDC30"     "PPIH"
"YBX1"       "CLDN19"     "P3H1"       "TMEM269"
 [529] "SVBP"       "ERMAP"      "ZNF691"     "SLC2A1"     "FAM183A"    "EBNA1BP2"   "CFAP57"
"TMEM125"    "C1orf210"   "TIE1"       "MPL"
 [540] "CDC20"      "ELOVL1"     "MED8"       "SZT2"       "HYI"        "PTPRF"      "KDM4A"
"ST3GAL3"    "ARTN"       "IPO13"      "DPH2"
 [551] "ATP6V0B"    "B4GALT2"    "CCDC24"     "SLC6A9"     "KLF17"      "KLF18"      "DMAP1"
"ERI3"       "RNF220"     "TMEM53"     "ARMH1"
 [562] "KIF2C"      "BEST4"      "PLK3"       "TCTEX1D4"   "BTBD19"     "PTCH2"      "EIF2B3"
"HECTD3"     "UROD"       "ZSWIM5"     "HPDL"
 [573] "MUTYH"      "TOE1"       "TESK2"      "CCDC163"    "MMACHC"     "PRDX1"      "AKR1A1"
"NASP"       "CCDC17"     "GPBP1L1"    "TMEM69"
 [584] "IPP"        "MAST2"      "PIK3R3"     "TSPAN1"     "P3R3URF"    "POMGNT1"    "LURAP1"
"RAD54L"     "LRRC41"     "UQCRH"      "NSUN4"
 [595] "FAAH"       "DMBX1"      "TMEM275"    "KNCN"       "MKNK1"      "MOB3C"      "ATPAF1"
"TEX38"      "EFCAB14"    "CYP4B1"     "CYP4A11"
 [606] "CYP4X1"     "CYP4Z1"     "CYP4A22"    "PDZK1IP1"   "TAL1"       "STIL"       "CMPK1"
```

```
"FOXE3"      "FOXD2"      "TRABD2B"    "SLC5A9"
 [617] "SPATA6"     "AGBL4"      "BEND5"      "ELAVL4"     "DMRTA2"     "FAF1"       "CDKN2C"
"C1orf185"   "RNF11"      "TTC39A"     "EPS15"
 [628] "OSBPL9"     "NRDC"       "RAB3B"      "TXNDC12"    "KTI12"      "BTF3L4"     "ZFYVE9"
"CC2D1B"     "ORC1"       "PRPF38A"    "TUT4"
 [639] "GPX7"       "SHISAL2A"   "COA7"       "ZYG11B"     "ZYG11A"     "ECHDC2"     "SCP2"
"PODN"       "SLC1A7"     "CPT2"       "CZIB"
 [650] "MAGOH"      "LRP8"       "DMRTB1"     "GLIS1"      "NDC1"       "YIPF1"      "DIO1"
"HSPB11"     "LRRC42"     "LDLRAD1"    "TMEM59"
 [661] "TCEANC2"    "CDCP2"      "CYB5RL"     "SSBP3"      "ACOT11"     "FAM151A"    "MROH7"
"TTC4"       "PARS2"      "TTC22"      "LEXM"
 [672] "DHCR24"     "TMEM61"     "BSND"       "PCSK9"      "USP24"      "PLPP3"      "PRKAA2"
"FYB2"       "C8A"        "C8B"        "DAB1"
 [683] "OMA1"       "TACSTD2"    "MYSM1"      "JUN"        "FGGY"       "HOOK1"      "CYP2J2"
"C1orf87"    "NFIA"       "TM2D1"      "PATJ"
 [694] "L1TD1"      "KANK4"      "USP1"       "DOCK7"      "ANGPTL3"    "ATG4C"      "FOXD3"
"ALG6"       "ITGB3BP"    "EFCAB7"     "PGM1"
 [705] "ROR1"       "UBE2U"      "CACHD1"     "RAVER2"     "JAK1"       "AK4"        "DNAJC6"
"LEPROT"     "LEPR"       "PDE4B"      "SGIP1"
 [716] "TCTEX1D1"   "INSL5"      "WDR78"      "MIER1"      "SLC35D1"    "C1orf141"   "IL23R"
"IL12RB2"    "SERBP1"     "GADD45A"    "GNG12"
 [727] "DIRAS3"     "WLS"        "RPE65"      "DEPDC1"     "LRRC7"      "LRRC40"     "SRSF11"
"ANKRD13C"   "HHLA3"      "CTH"        "PTGER3"
 [738] "ZRANB2"     "NEGR1"      "LRRIQ3"     "FPGT"       "TNNI3K"     "LRRC53"     "ERICH3"
"CRYZ"       "TYW3"       "LHX8"       "SLC44A5"
 [749] "ACADM"      "RABGGTB"    "MSH4"       "ASB17"      "ST6GALNAC3" "ST6GALNAC5" "PIGK"
"AK5"        "AC118549.1" "USP33"      "MIGA1"
 [760] "NEXN"       "FUBP1"      "DNAJB4"     "GIPC2"      "PTGFR"      "IFI44L"     "IFI44"
"ADGRL4"     "ADGRL2"     "TTLL7"      "PRKACB"
 [771] "SAMD13"     "DNASE2B"    "RPF1"       "GNG5"       "SPATA1"     "CTBS"       "SSX2IP"
"LPAR3"      "MCOLN2"     "WDR63"      "MCOLN3"
 [782] "SYDE2"      "C1orf52"    "BCL10"      "DDAH1"      "CCN1"       "ZNHIT6"     "COL24A
1"    "ODF2L"      "CLCA2"      "CLCA1"      "CLCA4"
 [793] "SH3GLB1"    "SELENOF"    "HS2ST1"     "LMO4"       "PKN2"       "GTF2B"      "KYAT3"
"RBMXL1"     "GBP3"       "GBP1"       "GBP2"
 [804] "GBP7"       "GBP4"       "GBP5"       "GBP6"       "LRRC8B"     "LRRC8C"     "LRRC8D"
"ZNF326"     "BARHL2"     "ZNF644"     "HFM1"
 [815] "CDC7"       "TGFBR3"     "BRDT"       "EPHX4"      "BTBD8"      "C1orf146"   "GLMN"
"RPAP2"      "GFI1"       "EVI5"       "DIPK1A"
 [826] "MTF2"       "TMED5"      "CCDC18"     "DR1"        "FNBP1L"     "BCAR3"      "DNTTIP
2"    "GCLM"       "ABCA4"      "ARHGAP29"   "ABCD3"
 [837] "F3"         "SLC44A3"    "CNN3"       "ALG14"      "TLCD4"      "RWDD3"      "PTBP2"
"DPYD"       "SNX7"       "PLPPR5"     "PLPPR4"
 [848] "PALMD"      "FRRS1"      "AGL"        "SLC35A3"    "MFSD14A"    "SASS6"      "TRMT13"
"LRRC39"     "DBT"        "RTCA"       "CDC14A"
 [859] "GPR88"      "VCAM1"      "EXTL2"      "SLC30A7"    "DPH5"       "S1PR1"      "OLFM3"
"COL11A1"    "RNPC3"      "AMY2A"      "PRMT6"
 [870] "NTNG1"      "VAV3"       "SLC25A24"   "FAM102B"    "HENMT1"     "PRPF38B"    "FNDC7"
"STXBP3"     "GPSM2"      "CLCC1"      "WDR47"
 [881] "TAF13"      "TMEM167B"   "C1orf194"   "KIAA1324"   "SARS"       "CELSR2"     "PSRC1"
"MYBPHL"     "SORT1"      "PSMA5"      "SYPL2"
 [892] "ATXN7L2"    "CYB561D1"   "AMIGO1"     "GPR61"      "GNAI3"      "GNAT2"      "AMPD2"
"GSTM1"      "GSTM3"      "EPS8L3"     "CSF1"
 [903] "AHCYL1"     "STRIP1"     "ALX3"       "UBL4B"      "SLC6A17"    "KCNC4"      "RBM15"
"SLC16A4"    "LAMTOR5"    "PROK1"      "KCNA10"
 [914] "KCNA2"      "KCNA3"      "CD53"       "LRIF1"      "DRAM2"      "CEPT1"      "DENND2
```

```
D"    "CHI3L2"     "CHIA"       "PIFO"       "OVGP1"
 [925] "WDR77"      "ATP5PB"     "C1orf162"   "TMIGD3"     "ADORA3"     "RAP1A"      "INKA2"
"DDX20"      "KCND3"      "CTTNBP2NL"  "WNT2B"
 [936] "ST7L"       "CAPZA1"     "MOV10"      "RHOC"       "PPM1J"      "TAFA3"      "SLC16A
1"    "LRIG2"      "MAGI3"      "PHTF1"      "RSBN1"
 [947] "PTPN22"     "BCL2L15"    "AP4B1"      "DCLRE1B"    "HIPK1"      "OLFML3"     "SYT6"
"TRIM33"     "BCAS2"      "DENND2C"    "AMPD1"
 [958] "NRAS"       "CSDE1"      "SIKE1"      "SYCP1"      "TSHB"       "TSPAN2"     "NGF"
"VANGL1"     "CASQ2"      "NHLH2"      "SLC22A15"
 [969] "MAB21L3"    "ATP1A1"     "CD58"       "IGSF3"      "CD2"        "PTGFRN"     "CD101"
"TTF2"       "TRIM45"     "VTCN1"      "MAN1A2"
 [980] "TENT5C"     "GDAP2"      "WDR3"       "SPAG17"     "TBX15"      "WARS2"      "HAO2"
"HSD3B2"     "HSD3B1"     "ZNF697"     "PHGDH"
 [991] "HMGCS2"     "REG4"       "ADAM30"     "NOTCH2"     "SEC22B"     "PPIAL4A"    "FCGR1B"
"NBPF20"     "PDZK1"      "CD160"
 [ reached 'max' / getOption("max.print") -- omitted 17071 entries ]
```

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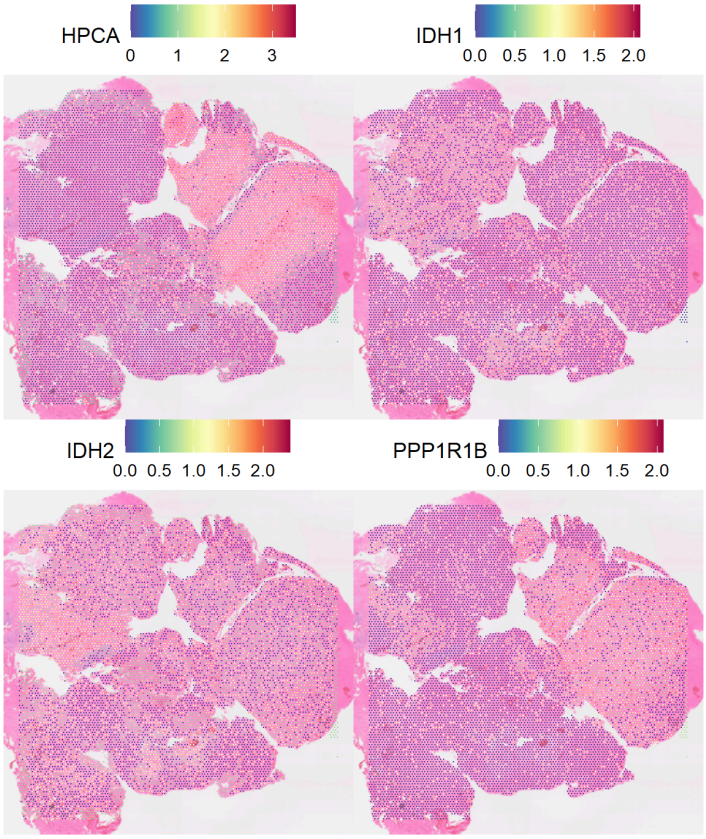
```
## Sanity check
'ATG5' %in% features
```

```
[1] TRUE
```

Visualisation of 4 Genes

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```
SpatialFeaturePlot(GBM, features = c("HPCA", 'IDH1', 'IDH2', 'PPP1R1B'))
```



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```
SpatialFeaturePlot(
  GBM,
  features = c('IDH1'),
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 2,
  image.scale = "lowres",
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
)
```



Dimensionality reduction, clustering

Hide

```
GBM <- RunPCA(GBM, assay = "SCT", verbose = FALSE)
GBM <- FindNeighbors(GBM, reduction = "pca", dims = 1:30)
GBM <- FindClusters(GBM, verbose = FALSE)
GBM <- RunUMAP(GBM, reduction = "pca", dims = 1:30)
```

```
Using method 'umap'
0%   10   20   30   40   50   60   70   80   90   100%
[----|----|----|----|----|----|----|----|----|----|
**************************************************|
```

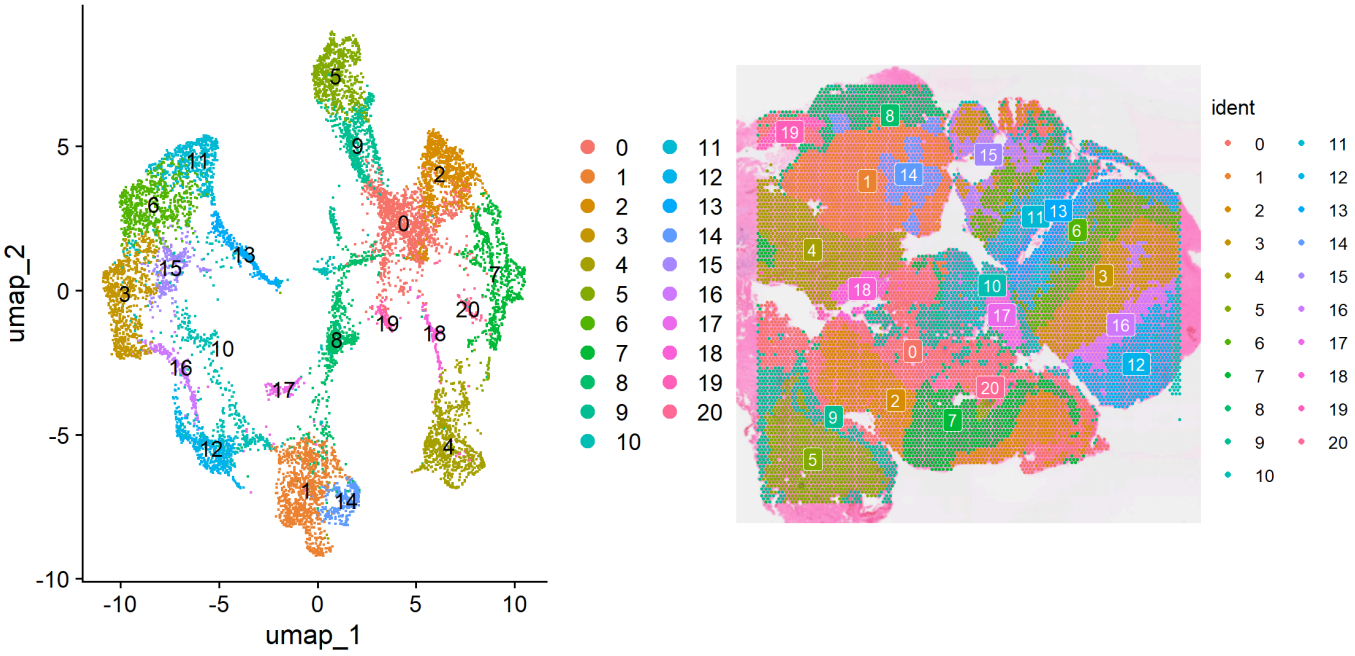
Visualisation

Hide

```
p1 <- DimPlot(GBM, reduction = "umap", label = TRUE)

p2 <- SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
)

p1 + p2
```



Searching for features exhibiting spatial patterning in the absence of pre-annotation

Using two methods

Moran's I Algorithm

Hide

```
GBM <- FindSpatiallyVariableFeatures(GBM,
                                     features = VariableFeatures(GBM)[1:1000],
                                     selection.method = "moransi")
```

```
|                                                                      | 0 % ~calculating
|+                                                                     | 1 % ~11m 05s
|+                                                                     | 2 % ~10m 30s
|++                                                                    | 3 % ~09m 57s
|++                                                                    | 4 % ~09m 21s
|+++                                                                   | 5 % ~08m 56s
|+++                                                                   | 6 % ~08m 37s
|++++                                                                  | 7 % ~08m 22s
|++++                                                                  | 8 % ~08m 08s
|+++++                                                                 | 9 % ~08m 02s
|+++++                                                                 | 10% ~07m 56s
|++++++                                                                | 11% ~07m 50s
|++++++                                                                | 12% ~07m 42s
|+++++++                                                               | 13% ~07m 33s
|+++++++                                                               | 14% ~07m 23s
|++++++++                                                              | 15% ~07m 19s
|++++++++                                                              | 16% ~07m 15s
|+++++++++                                                             | 17% ~07m 12s
|+++++++++                                                             | 18% ~07m 08s
|++++++++++                                                            | 19% ~07m 03s
|++++++++++                                                            | 20% ~06m 57s
|+++++++++++                                                           | 21% ~06m 51s
|+++++++++++                                                           | 22% ~06m 45s
|++++++++++++                                                          | 23% ~06m 40s
|++++++++++++                                                          | 24% ~06m 35s
|+++++++++++++                                                         | 25% ~06m 30s
|+++++++++++++                                                         | 26% ~06m 25s
|++++++++++++++                                                        | 27% ~06m 20s
|++++++++++++++                                                        | 28% ~06m 15s
|+++++++++++++++                                                       | 29% ~06m 12s
|+++++++++++++++                                                       | 30% ~06m 08s
|++++++++++++++++                                                      | 31% ~06m 03s
|++++++++++++++++                                                      | 32% ~05m 60s
|+++++++++++++++++                                                     | 33% ~05m 55s
|+++++++++++++++++                                                     | 34% ~05m 50s
|++++++++++++++++++                                                    | 35% ~05m 44s
|++++++++++++++++++                                                    | 36% ~05m 39s
|+++++++++++++++++++                                                   | 37% ~05m 34s
|+++++++++++++++++++                                                   | 38% ~05m 29s
|++++++++++++++++++++                                                  | 39% ~05m 24s
|++++++++++++++++++++                                                  | 40% ~05m 19s
|+++++++++++++++++++++                                                 | 41% ~05m 14s
|+++++++++++++++++++++                                                 | 42% ~05m 09s
|++++++++++++++++++++++                                                | 43% ~05m 04s
|++++++++++++++++++++++                                                | 44% ~04m 59s
|+++++++++++++++++++++++                                               | 45% ~04m 54s
|+++++++++++++++++++++++                                               | 46% ~04m 48s
|++++++++++++++++++++++++                                              | 47% ~04m 43s
|++++++++++++++++++++++++                                              | 48% ~04m 38s
|+++++++++++++++++++++++++                                             | 49% ~04m 33s
|+++++++++++++++++++++++++                                             | 50% ~04m 28s
|++++++++++++++++++++++++++                                            | 51% ~04m 23s
|++++++++++++++++++++++++++                                            | 52% ~04m 17s
|+++++++++++++++++++++++++++                                           | 53% ~04m 12s
```

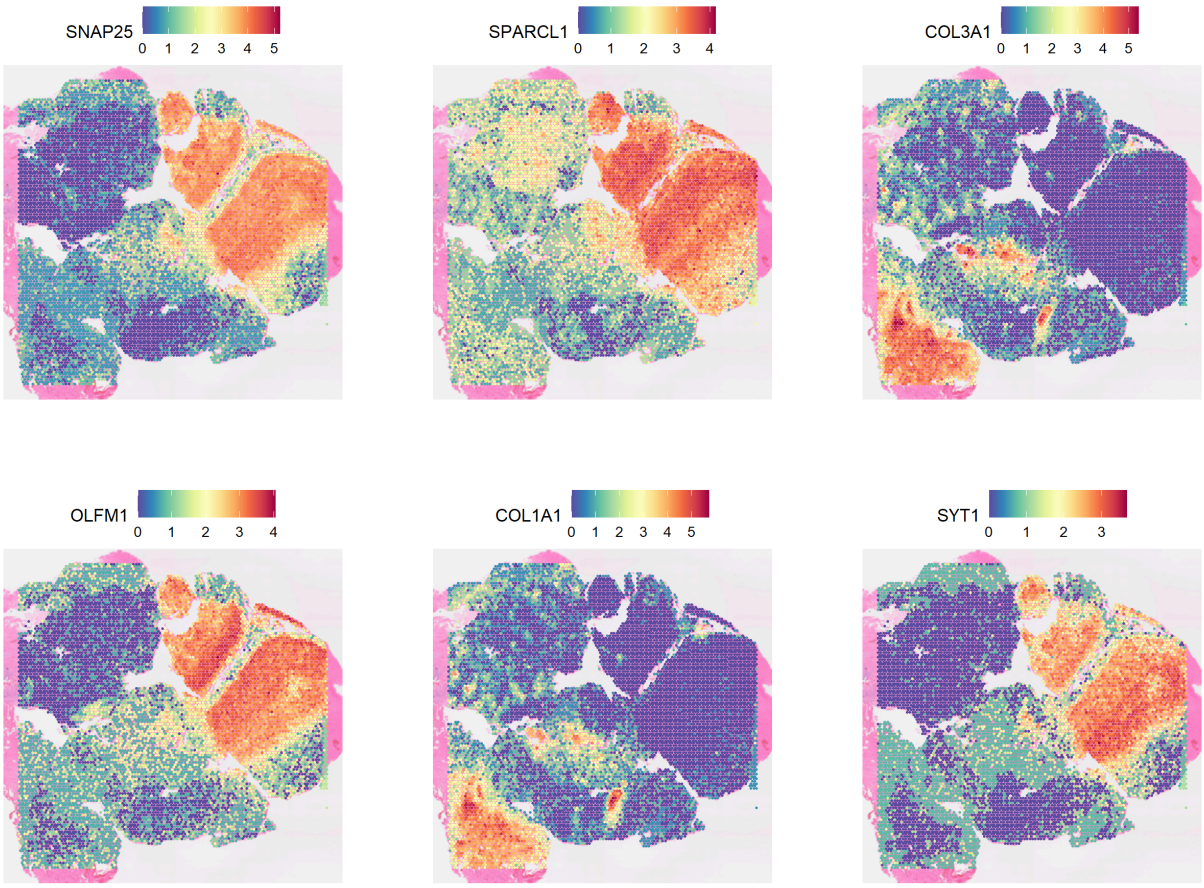
```
|++++++++++++++++++++++++++                          | 54% ~04m 06s
|+++++++++++++++++++++++++++                         | 55% ~04m 01s
|++++++++++++++++++++++++++++                        | 56% ~03m 55s
|++++++++++++++++++++++++++++                        | 57% ~03m 50s
|++++++++++++++++++++++++++++                        | 58% ~03m 44s
|+++++++++++++++++++++++++++++                       | 59% ~03m 40s
|+++++++++++++++++++++++++++++                       | 60% ~03m 34s
|+++++++++++++++++++++++++++++                       | 61% ~03m 29s
|++++++++++++++++++++++++++++++                      | 62% ~03m 23s
|++++++++++++++++++++++++++++++                      | 63% ~03m 18s
|++++++++++++++++++++++++++++++                      | 64% ~03m 12s
|+++++++++++++++++++++++++++++++                     | 65% ~03m 07s
|+++++++++++++++++++++++++++++++                     | 66% ~03m 01s
|+++++++++++++++++++++++++++++++                     | 67% ~02m 56s
|++++++++++++++++++++++++++++++++                    | 68% ~02m 50s
|++++++++++++++++++++++++++++++++                    | 69% ~02m 45s
|++++++++++++++++++++++++++++++++                    | 70% ~02m 40s
|+++++++++++++++++++++++++++++++++                   | 71% ~02m 34s
|+++++++++++++++++++++++++++++++++                   | 72% ~02m 29s
|+++++++++++++++++++++++++++++++++                   | 73% ~02m 24s
|++++++++++++++++++++++++++++++++++                  | 74% ~02m 18s
|++++++++++++++++++++++++++++++++++                  | 75% ~02m 13s
|++++++++++++++++++++++++++++++++++                  | 76% ~02m 08s
|+++++++++++++++++++++++++++++++++++                 | 77% ~02m 03s
|+++++++++++++++++++++++++++++++++++                 | 78% ~01m 57s
|+++++++++++++++++++++++++++++++++++                 | 79% ~01m 52s
|++++++++++++++++++++++++++++++++++++                | 80% ~01m 47s
|++++++++++++++++++++++++++++++++++++                | 81% ~01m 41s
|++++++++++++++++++++++++++++++++++++                | 82% ~01m 36s
|+++++++++++++++++++++++++++++++++++++               | 83% ~01m 31s
|+++++++++++++++++++++++++++++++++++++               | 84% ~01m 25s
|+++++++++++++++++++++++++++++++++++++               | 85% ~01m 20s
|++++++++++++++++++++++++++++++++++++++              | 86% ~01m 15s
|++++++++++++++++++++++++++++++++++++++              | 87% ~01m 09s
|++++++++++++++++++++++++++++++++++++++              | 88% ~01m 04s
|+++++++++++++++++++++++++++++++++++++++             | 89% ~58s
|+++++++++++++++++++++++++++++++++++++++             | 90% ~53s
|+++++++++++++++++++++++++++++++++++++++             | 91% ~48s
|++++++++++++++++++++++++++++++++++++++++            | 92% ~43s
|++++++++++++++++++++++++++++++++++++++++            | 93% ~37s
|++++++++++++++++++++++++++++++++++++++++            | 94% ~32s
|+++++++++++++++++++++++++++++++++++++++++           | 95% ~27s
|+++++++++++++++++++++++++++++++++++++++++           | 96% ~21s
|++++++++++++++++++++++++++++++++++++++++++          | 97% ~16s
|++++++++++++++++++++++++++++++++++++++++++          | 98% ~11s
|++++++++++++++++++++++++++++++++++++++++++          | 99% ~05s
|++++++++++++++++++++++++++++++++++++++++++          | 100% elapsed=08m 51s
```

Hide

```
# With data

top.features <- head(SpatiallyVariableFeatures(GBM, method = "moransi"), 6)

SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```




```
```

Interestingly, Clusters 15, 11, 13, 6, 3, 16, 12 have a common base of overly expressed group of genes (could they be actually way more similar than expected and therefore PCA is overclustering?)

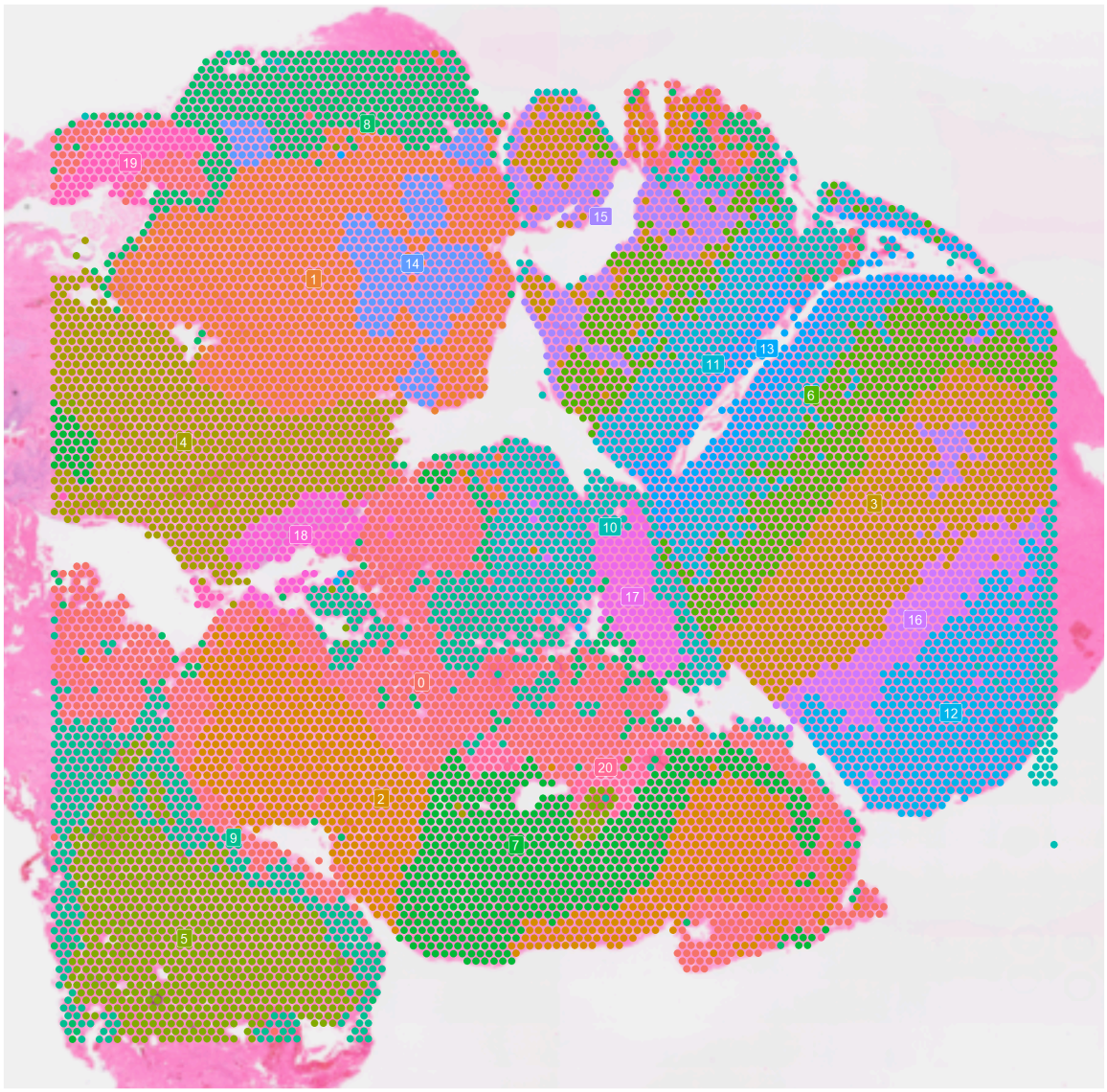
```
```

Hide

```
# With scale data

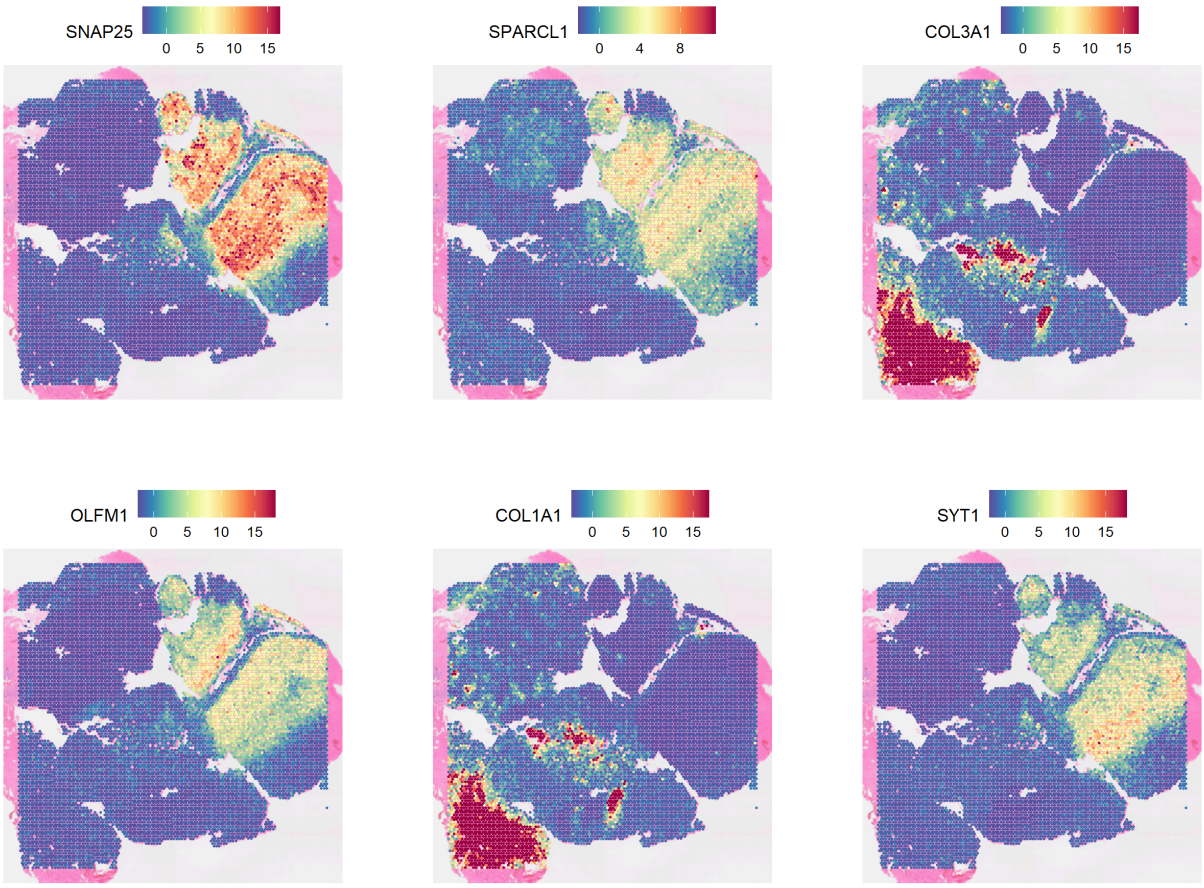
top.features <- head(SpatiallyVariableFeatures(GBM, method = "moransi"), 6)

SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
) +
  NoLegend()
```



Hide

```
SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "scale.data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```



Mark Variogram Algorithm

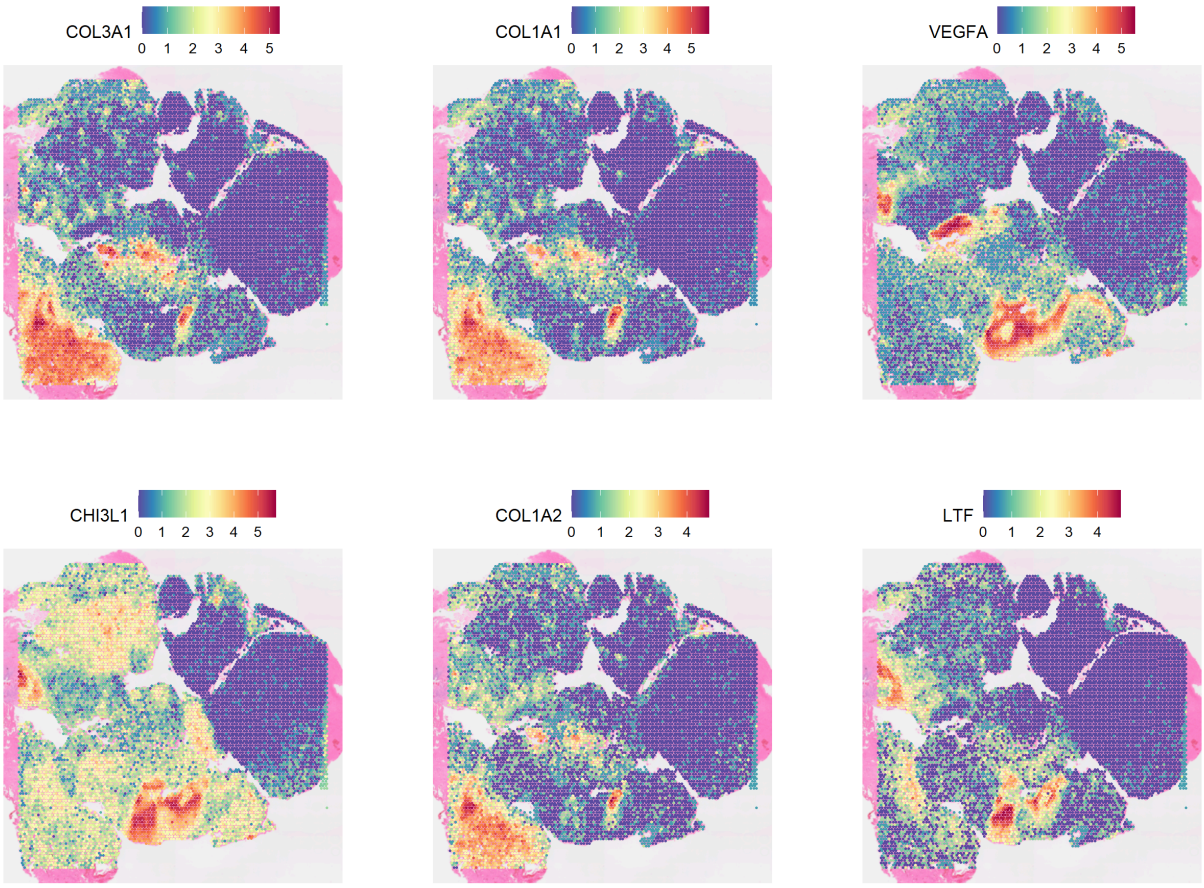
Hide

```
GBM <- FindSpatiallyVariableFeatures(GBM,
                                     features = VariableFeatures(GBM)[1:10],
                                     selection.method = "markvariogram")
```

Hide

```
# With data
top.features <- head(SpatiallyVariableFeatures(GBM, method = "markvariogram"), 6)

SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```



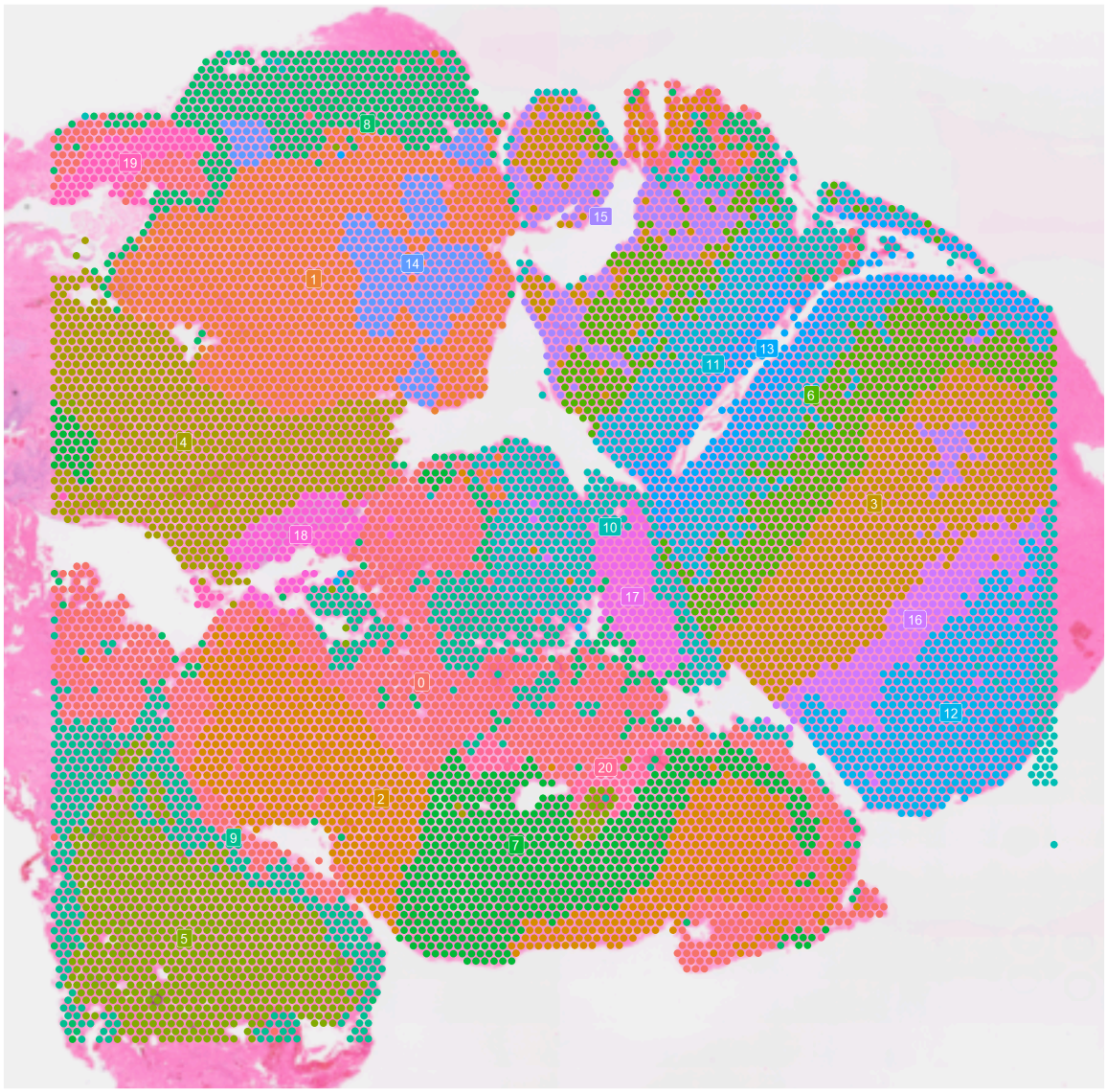
The patterns of Spatially Variable Features most certainly changed between algorythms, I will be moving forward with

‘COL’ type genes seem to be common to a certain specific region of the brain with both algorythms.

Hide

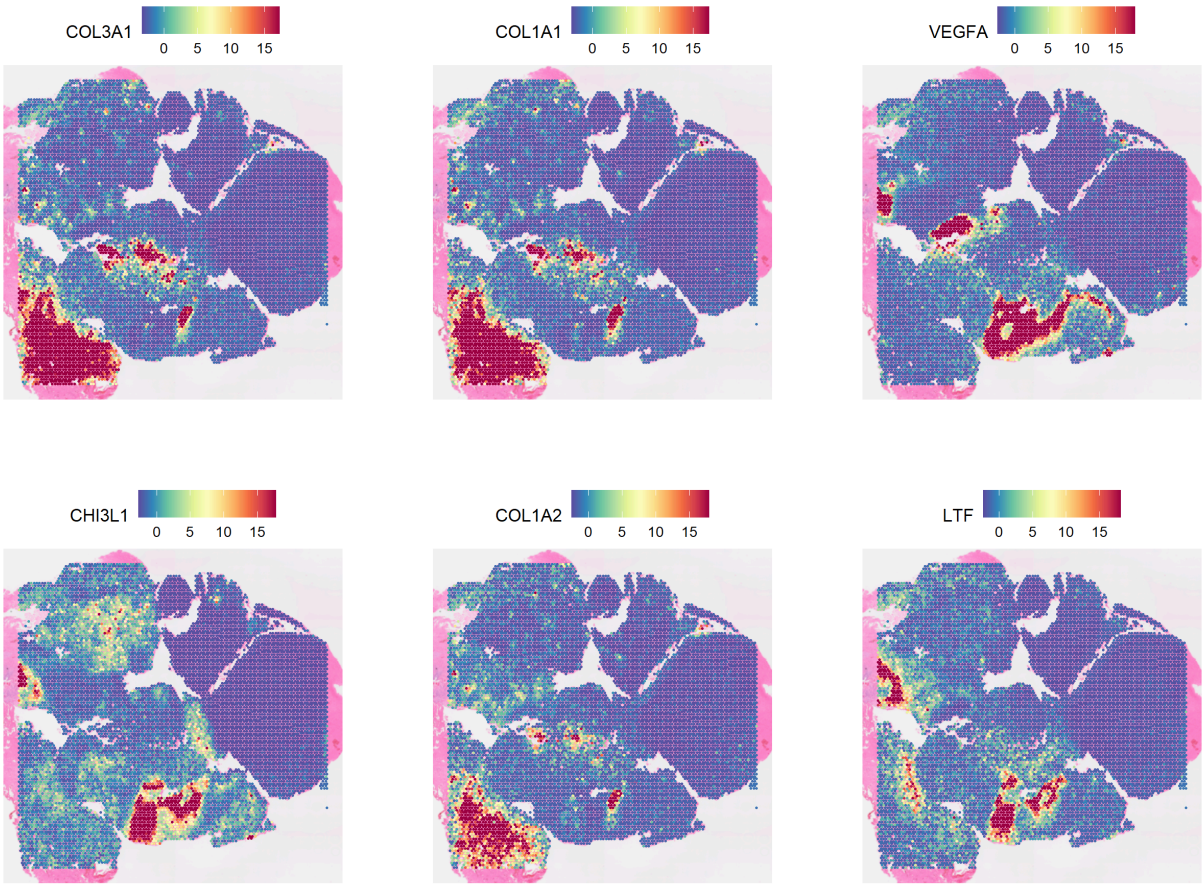
```
# With scale data
top.features <- head(SpatiallyVariableFeatures(GBM, method = "markvariogram"), 6)

SpatialDimPlot(
  GBM,
  group.by = NULL,
  images = NULL,
  cols = NULL,
  crop = TRUE,
  cells.highlight = NULL,
  cols.highlight = c("#DE2D26", "grey50"),
  facet.highlight = TRUE,
  label = TRUE,
  label.size = 3,
  label.color = "white",
  repel = FALSE,
  ncol = NULL,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 1,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  label.box = TRUE,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = FALSE
) +
  NoLegend()
```



Hide

```
SpatialFeaturePlot(
  GBM,
  features = top.features,
  images = NULL,
  crop = TRUE,
  slot = "scale.data",
  keep.scale = "feature",
  min.cutoff = NA,
  max.cutoff = NA,
  ncol = 3,
  combine = TRUE,
  pt.size.factor = 2.5,
  alpha = c(2, 2),
  image.alpha = 2,
  image.scale = "lowres",
  shape = 21,
  stroke = NA,
  stroke.alpha = NA,
  interactive = FALSE,
  information = NULL,
  plot_segmentations = TRUE
) &
  theme(plot.margin = unit(c(1, 1, 1, 1), "cm"))
```



Inspecting clusters and finding their TOP markers to study patterns

Hide

```
table(Idents(GBM))
```

```
   0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17   1
8   19   20
1262  969  902  795  757  725  677  649  616  571  557  482  478  307  272  255  214  124  11
2   91   63
```

Hide

```
markers <- FindAllMarkers(
  GBM,
  only.pos = TRUE,
  min.pct = 0.25,
  logfc.threshold = 0.25
)
```

10 Top Genes per cluster

Hide

```
library(dplyr)

top10 <- markers %>%
  group_by(cluster) %>%
  slice_max(
    order_by = avg_log2FC,
    n = 10
  )

top10
```

p_val <dbl>	avg_log2FC <dbl>	pct.1 <dbl>	pct.2 <dbl>	p_val_adj <dbl>	cluster <fctr>	gene <chr>
6.937356e-280	2.0979328	0.666	0.246	1.253650e-275	0	IL26
1.251727e-243	2.0844889	0.615	0.229	2.261996e-239	0	IL22
2.726333e-118	1.5930694	0.563	0.333	4.926756e-114	0	NUP107
1.885771e-40	1.4564387	0.254	0.130	3.407777e-36	0	MUC12
6.712439e-111	1.4468966	0.605	0.407	1.213005e-106	0	SZRD1
1.145808e-35	1.3234250	0.268	0.147	2.070589e-31	0	FLG
7.053281e-239	1.2664401	0.999	0.833	1.274598e-234	0	CAND1
2.711153e-89	1.2374792	0.539	0.309	4.899325e-85	0	ELOA3
7.295363e-97	1.2373416	0.575	0.345	1.318345e-92	0	FAM90A1
2.490170e-164	1.2184393	0.805	0.438	4.499986e-160	0	E2F3

1-10 of 210 rows Previous **1** 2 3 4 5 6 … 21 Next

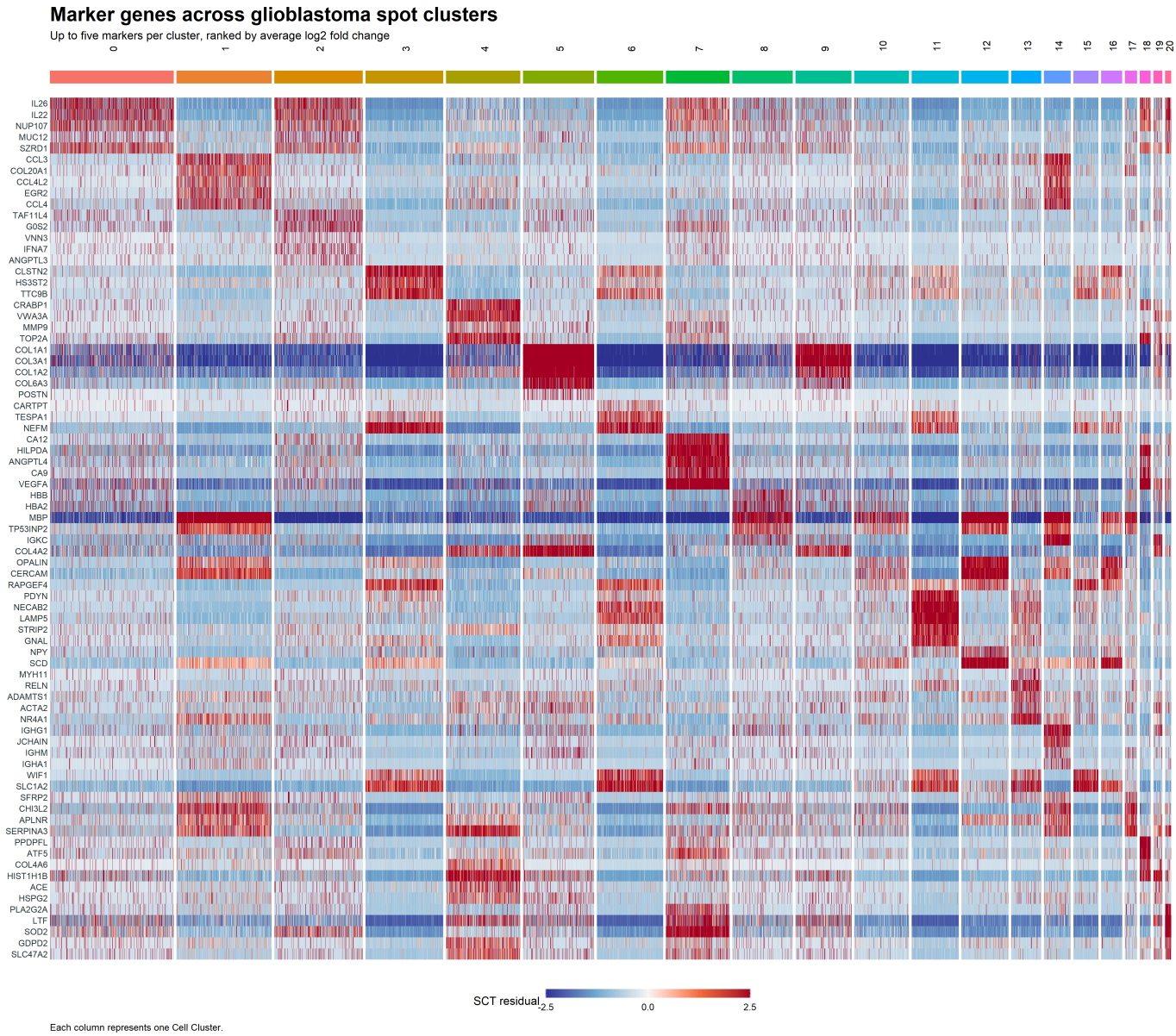
Hide

```
top5 <- top10 %>%
  group_by(cluster) %>%
  slice_max(order_by = avg_log2FC, n = 5, with_ties = FALSE) %>%
  ungroup()

residual_limit <- 2.5

p_heatmap <- DoHeatmap(
  GBM,
  features = unique(top5$gene),
  assay = "SCT",
  slot = "scale.data",
  group.by = "seurat_clusters",
  disp.min = -residual_limit,
  disp.max = residual_limit,
  angle = 90,
  hjust = 0,
  vjust = 0.5,
  size = 3.4,
  group.bar.height = 0.015,
  combine = FALSE
)[[1]] +
  scale_fill_gradientn(
    colours = c("#313695", "#74ADD1", "#F7F7F7", "#F46D43", "#A50026"),
    limits = c(-residual_limit, residual_limit),
    breaks = c(-residual_limit, 0, residual_limit),
    na.value = "white",
    name = "SCT residual"
  ) +
  labs(
    title = "Marker genes across glioblastoma spot clusters",
    subtitle = "Up to five markers per cluster, ranked by average log2 fold change",
    caption = "Each column represents one Cell Cluster."
  ) +
  theme(
    axis.text.y = element_text(size = 8.5, colour = "#263238"),
    legend.position = "bottom",
    legend.key.width = grid::unit(14, "mm"),
    legend.key.height = grid::unit(3, "mm"),
    plot.title = element_text(size = 19, face = "bold"),
    plot.subtitle = element_text(size = 11, margin = margin(b = 10)),
    plot.caption = element_text(size = 9, hjust = 0),
    plot.margin = margin(8, 12, 6, 8)
  )

p_heatmap + ggplot2::guides(colour = "none")
```



Possible correlation

Identification of Spatially Variable Features using Seurat

Perform differential expression based on pre-annotated anatomical regions within the tissue, which may be determined either from unsupervised clustering or prior knowledge.

Cluster 5 and 9 have ‘COL’ type genes overly expressed when compared to all other clusters

Hide

```
avg_sct_residuals <- as.matrix(
  AverageExpression(
    GBM,
    assays = "SCT",
    group.by = "seurat_clusters",
    layer = "scale.data"
  )$SCT
)

rho <- cor(avg_sct_residuals, method = "spearman")
cor_5_9 <- rho["g5", "g9"]

matches_5 <- sort(
  rho["g5", colnames(rho) != "g5"],
  decreasing = TRUE
)

data.frame(
  rank = seq_along(matches_5),
  cluster = sub("^g", "", names(matches_5)),
  spearman_rho = round(unname(matches_5), 3)
)
```

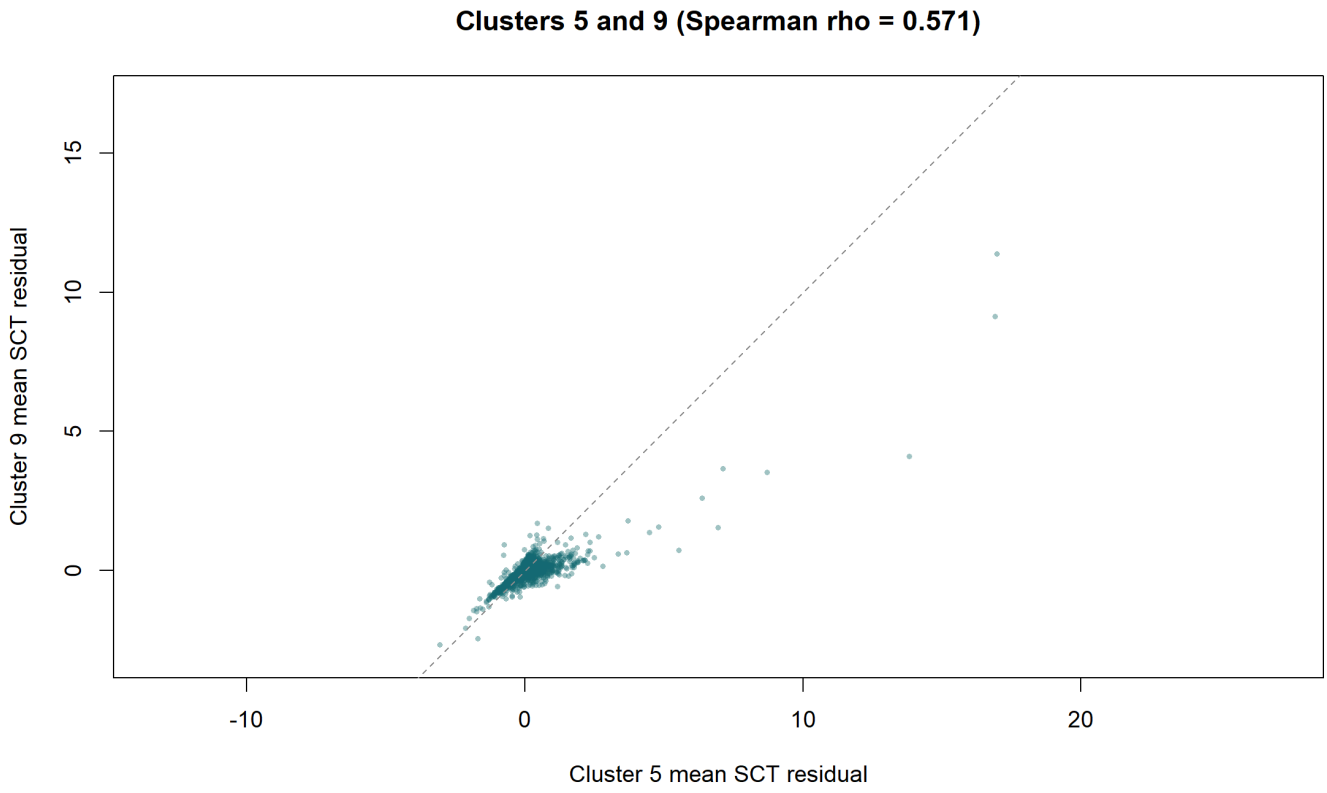
rank	cluster	spearman_rho
<int>	<chr>	<dbl>
1	9	0.571
2	19	0.540
3	4	0.483
4	14	0.447
5	8	0.435
6	1	0.420
7	20	0.342
8	7	0.335
9	0	0.283
10	2	0.240

1-10 of 20 rows Previous **1** 2 Next

Hide

```
pair <- avg_sct_residuals[, c("g5", "g9")]
limits <- range(pair)

plot(
  pair[, "g5"], pair[, "g9"],
  xlim = limits, ylim = limits, asp = 1,
  pch = 16, cex = 0.55,
  col = adjustcolor("#176C72", alpha.f = 0.4),
  xlab = "Cluster 5 mean SCT residual",
  ylab = "Cluster 9 mean SCT residual",
  main = sprintf("Clusters 5 and 9 (Spearman rho = %.3f)", cor_5_9)
)
abline(a = 0, b = 1, lty = 2, col = "grey55")
```



Mapping the clusters and gene programms

This highlights clusters 5 (Collagen), 9 (Collagen) and 7 (Hypoxia) on the same H&E image.

Hide

```
focus <- c("5", "9", "7")
cluster <- as.character(GBM$seurat_clusters)
GBM$focus_region <- factor(
  ifelse(cluster %in% focus, cluster, "Other"),
  levels = c(focus, "Other")
)

p_clusters <- SpatialDimPlot(
  GBM,
  group.by = "focus_region",
  pt.size.factor = 2.8,
  alpha = 1,
  image.alpha = 0.55,
  cols = c(
    "5" = "#D55E00",
    "9" = "#009E73",
    "7" = "#0072B2",
    "Other" = "grey88"
  ),
  combine = FALSE
)[[1]]

panels <- list(
  Collagen = c("COL1A1", "COL1A2", "COL3A1", "COL6A3"),
  GBM_markers = c("BCAN", "PTPRZ1", "DDR1"),
  Hypoxia = c("CA12", "HILPDA", "ANGPTL4", "CA9", "VEGFA"),
  MES1_markers = c("CHI3L1", "CD44", "ANXA2"),
  MES2_markers = c("ADM", "LDHA", "DDIT3")
)

panels <- lapply(panels, intersect, y = rownames(GBM[["SCT"]]))
print(panels)
```

```
$Collagen
[1] "COL1A1" "COL1A2" "COL3A1" "COL6A3"

$GBM_markers
[1] "BCAN"   "PTPRZ1" "DDR1"

$Hypoxia
[1] "CA12"    "HILPDA"  "ANGPTL4" "CA9"     "VEGFA"

$MES1_markers
[1] "CHI3L1" "CD44"   "ANXA2"

$MES2_markers
[1] "ADM"   "DDIT3"
```

Hide

```
GBM <- AddModuleScore(
  GBM, features = panels, assay = "SCT", name = "Niche", seed = 42
)
GBM$Collagen <- GBM$Niche1
GBM$GBM_markers <- GBM$Niche2
GBM$Hypoxia <- GBM$Niche3
GBM$MES1_markers <- GBM$Niche4
GBM$MES2_markers <- GBM$Niche5

score_maps <- SpatialFeaturePlot(
  GBM,
  features = c(
    "Collagen", "Hypoxia",
    "MES1_markers", "MES2_markers",
    "GBM_markers"
  ),
  pt.size.factor = 2.5,
  alpha = c(1, 1),
  image.alpha = 0.55,
  keep.scale = "feature",
  combine = FALSE
)
```

Hide

```
## Not a must, just making the graphs more readble
p_clusters <- p_clusters +
  ggplot2::guides(
    fill = ggplot2::guide_legend(
      override.aes = list(size = 6, alpha = 1)
    )
  )

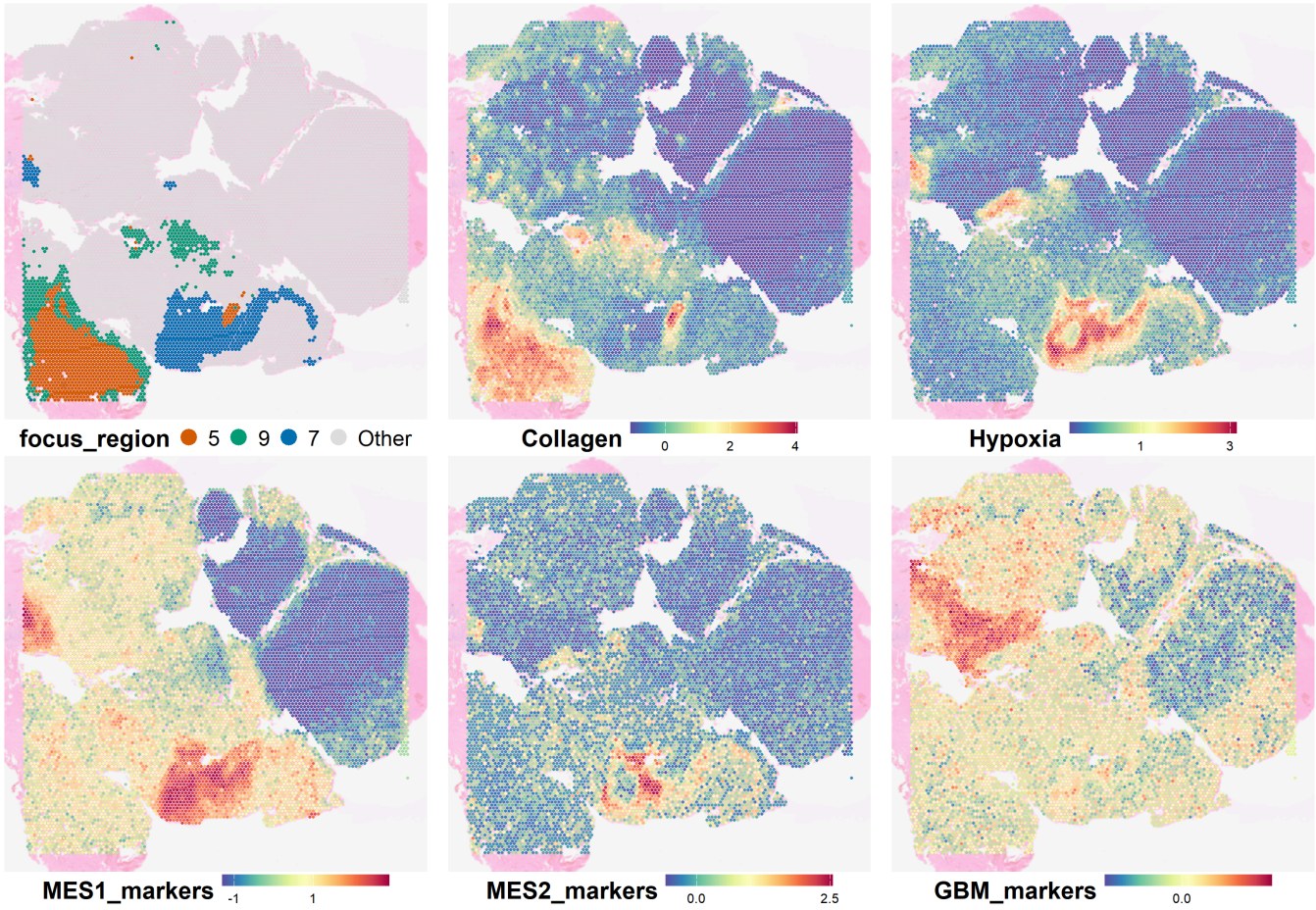
score_maps <- lapply(score_maps, function(p) {
  fill_scale <- p$scales$get_scales("fill")
  fill_scale$breaks <- scales::breaks_extended(n = 3)

  p + ggplot2::guides(
    fill = ggplot2::guide_colourbar(
      direction = "horizontal",
      theme = ggplot2::theme(
        legend.key.width = grid::unit(44, "mm"),
        legend.key.height = grid::unit(3, "mm"),
        legend.text = ggplot2::element_text(size = 10)
      )
    )
  )
})

## Plotting

p_spatial <- wrap_plots(
  c(list(p_clusters), score_maps),
  ncol = 3,
  nrow = 2,
  guides = "keep"
) & ggplot2::theme(
  plot.margin = ggplot2::margin(t = 1, r = 8, b = 1, l = 8, unit = "pt"),
  legend.position = "bottom",
  legend.box.spacing = grid::unit(2, "pt"),
  legend.margin = ggplot2::margin(0, 0, 0, 0),
  legend.text = ggplot2::element_text(size = 16),
  legend.title = ggplot2::element_text(size = 18, face = "bold")
)

p_spatial
```



Comparing Clusters highly expression MBP

Hide

```
avg_expr <- AverageExpression(
  GBM,
  assays = "SCT",
  group.by = "seurat_clusters",
  layer = "scale.data"
)$SCT

clusters_to_compare <- c(
  "g1", "g10", "g14", "g17",
  "g8", "g12", "g16"
)

avg_expr_subset <- as.matrix(
  avg_expr[, clusters_to_compare, drop = FALSE]
)

cor_matrix <- cor(
  avg_expr_subset,
  method = "spearman"
)

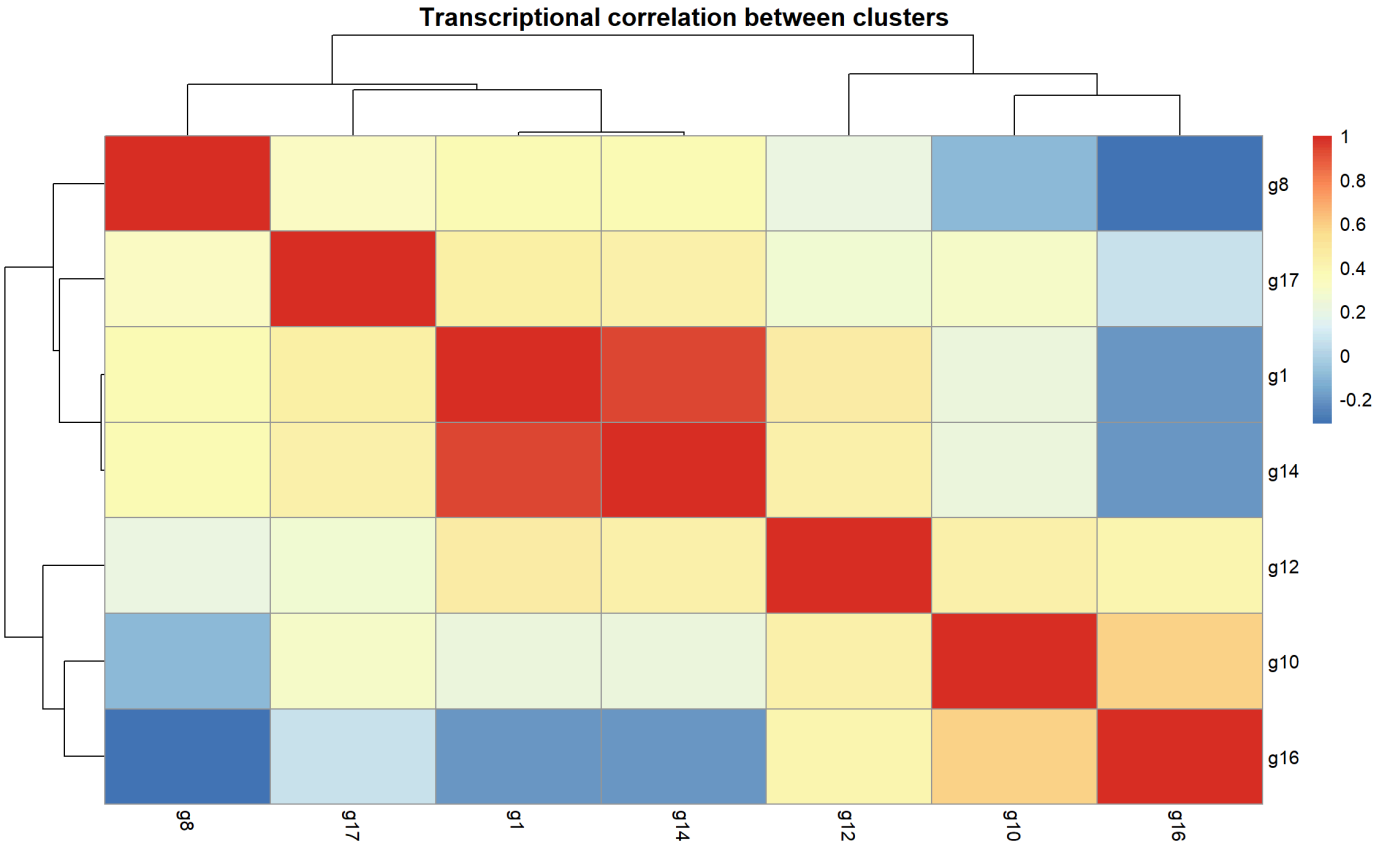
round(cor_matrix, 3)
```

```
        g1    g10    g14   g17     g8   g12    g16
g1   1.000  0.224  0.944 0.440  0.374 0.457 -0.194
g10  0.224  1.000  0.229 0.299 -0.090 0.431  0.587
g14  0.944  0.229  1.000 0.424  0.376 0.434 -0.202
g17  0.440  0.299  0.424 1.000  0.320 0.277  0.065
g8   0.374 -0.090  0.376 0.320  1.000 0.210 -0.312
g12  0.457  0.431  0.434 0.277  0.210 1.000  0.407
g16 -0.194  0.587 -0.202 0.065 -0.312 0.407  1.000
```

Hide

```
library(pheatmap)

pheatmap(
  cor_matrix,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  main = "Transcriptional correlation between clusters"
)
```



Hide

```
cor_pairs <- as.data.frame(as.table(cor_matrix))

cor_pairs <- cor_pairs %>%
  filter(Var1 != Var2) %>%
  mutate(
    pair = paste(
      pmin(as.character(Var1), as.character(Var2)),
      pmax(as.character(Var1), as.character(Var2)),
      sep = " vs "
    )
  ) %>%
  distinct(pair, .keep_all = TRUE) %>%
  arrange(desc(Freq))

cor_pairs
```

Var1 <fctr>	**Var2** <fctr>	**Freq** <dbl>	**pair** <chr>
g14	g1	0.94376386	g1 vs g14
g16	g10	0.58682826	g10 vs g16
g12	g1	0.45725566	g1 vs g12
g17	g1	0.43991782	g1 vs g17
g12	g14	0.43404108	g12 vs g14
g12	g10	0.43060405	g10 vs g12
g17	g14	0.42397366	g14 vs g17
g16	g12	0.40749440	g12 vs g16
g8	g14	0.37597663	g14 vs g8
g8	g1	0.37352477	g1 vs g8

1-10 of 21 rows Previous **1** 2 3 Next

Hide

```
library(Seurat)
library(dplyr)
library(ggplot2)
library(patchwork)

gbm_myelin <- GBM
DefaultAssay(gbm_myelin) <- "SCT"

support_genes <- c("PLP1", "MOG", "MOBP")
genes <- c("MBP", support_genes)

missing_genes <- setdiff(genes, rownames(gbm_myelin[["SCT"]]))
if (length(missing_genes) > 0) {
  stop("Genes missing from the SCT assay: ", paste(missing_genes, collapse = ", "))
}

cluster_ids <- as.character(gbm_myelin$seurat_clusters)
gbm_myelin$myelin_region <- factor(
  ifelse(cluster_ids %in% c("1", "14"), paste("Cluster", cluster_ids), "Other spots"),
  levels = c("Cluster 1", "Cluster 14", "Other spots")
)

gbm_myelin <- AddModuleScore(
  gbm_myelin,
  features = list(support_genes),
  assay = "SCT",
  name = "MyelinCheck",
  seed = 42
)

image_id <- Images(gbm_myelin)[1]

p_clusters <- SpatialDimPlot(
  gbm_myelin,
  group.by = "myelin_region",
  images = image_id,
  image.scale = "lowres",
  cols = c(
    "Cluster 1" = "#D55E00",
    "Cluster 14" = "#0072B2",
    "Other spots" = "#D9D9D9"
  ),
  pt.size.factor = 2.5,
  image.alpha = 0.55,
  combine = FALSE
)[[1]] + ggtitle("Clusters 1 and 14")

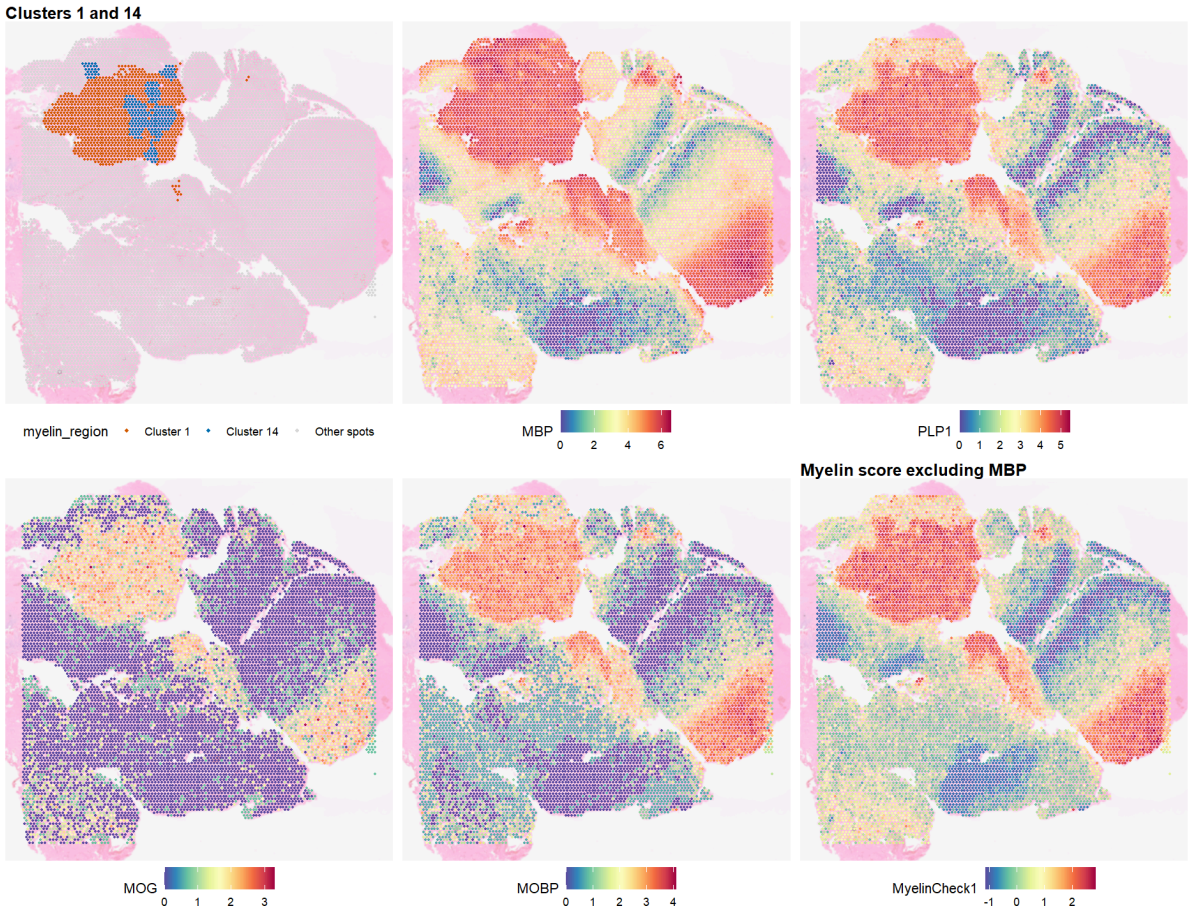
marker_maps <- SpatialFeaturePlot(
  gbm_myelin,
  features = c(genes, "MyelinCheck1"),
  slot = "data",
  images = image_id,
  image.scale = "lowres",
  pt.size.factor = 2.5,
  image.alpha = 0.55,
```

```
  keep.scale = "feature",
  combine = FALSE
)

score_panel <- length(marker_maps)
marker_maps[[score_panel]] <- marker_maps[[score_panel]] +
  ggtitle("Myelin score excluding MBP")

p_myelin <- wrap_plots(c(list(p_clusters), marker_maps), ncol = 3) &
  theme(
    plot.margin = margin(2, 4, 2, 4),
    legend.position = "bottom",
    plot.title = element_text(size = 13, face = "bold")
  )

p_myelin
```



Hide

```
spot_data <- FetchData(
  gbm_myelin,
  vars = c("myelin_region", genes, "MyelinCheck1"),
  layer = "data"
)

spot_data %>%
  group_by(myelin_region) %>%
  summarise(
    spots = n(),
    across(all_of(genes), mean, .names = "mean_{.col}"),
    median_myelin_score = median(MyelinCheck1),
    .groups = "drop"
  )
```

myelin_region <fctr>	sp… <int>	mean_… <dbl>	mean_P… <dbl>	mean_M… <dbl>	mean_M… <dbl>	median_myelin_score <dbl>
Cluster 1	969	5.496520	4.420018	1.8037202	2.963050	2.1170766
Cluster 14	272	5.442496	4.386883	1.7914450	2.936575	2.1202915
Other spots	9637	3.077577	1.917730	0.4151402	1.046296	0.1013383

3 rows

Hide

```
mbp_spots <- FetchData(
  gbm_myelin,
  vars = c("seurat_clusters", "MBP", "MyelinCheck1", "nCount_Spatial"),
  layer = "data"
)

focus <- c("1", "14")
mbp_cutoff <- quantile(mbp_spots$MBP, 0.80, names = FALSE)

mbp_spots$cluster <- as.character(mbp_spots$seurat_clusters)
mbp_spots$mbp_high <- mbp_spots$MBP > 0 & mbp_spots$MBP >= mbp_cutoff

mbp_summary <- mbp_spots %>%
  group_by(cluster) %>%
  summarise(
    spots = n(),
    high_spots = sum(mbp_high),
    pct_high = round(100 * mean(mbp_high), 1),
    median_MBP_high = median(MBP[mbp_high]),
    median_myelin_high = median(MyelinCheck1[mbp_high]),
    median_UMIs_high = median(nCount_Spatial[mbp_high]),
    .groups = "drop"
  ) %>%
  arrange(desc(pct_high))

print(mbp_summary, n = Inf)
```

```
# A tibble: 21 × 7
   cluster spots high_spots pct_high median_MBP_high median_myelin_high median_UMIs_high
   <chr>   <int>      <int>    <dbl>           <dbl>              <dbl>            <dbl>
 1 12        478        456     95.4            5.68               2.13             8796.
 2 1         969        891     92              5.52               2.13            15237
 3 14        272        236     86.8            5.45               2.14            13759
 4 8         616        343     55.7            5.58               1.21             2376
 5 16        214         72     33.6            5.39               1.79            11532.
 6 10        557        152     27.3            5.49               1.93             6103
 7 17        124         13     10.5            5.32               1.66             7006
 8 18        112          3      2.7            5.62               0.822            3174
 9 19         91          2      2.2            5.27               0.953            4998
10 4         757          5      0.7            5.48               1.29            12384
11 0        1262          8      0.6            5.35               1.04             3468.
12 13        307          1      0.3            5.64               0.944            3232
13 5         725          1      0.1            5.33               1.43             2831
14 11        482          0      0             NA                 NA                NA
15 15        255          0      0             NA                 NA                NA
16 2         902          0      0             NA                 NA                NA
17 20         63          0      0             NA                 NA                NA
18 3         795          0      0             NA                 NA                NA
19 6         677          0      0             NA                 NA                NA
20 7         649          0      0             NA                 NA                NA
21 9         571          0      0             NA                 NA                NA
```

Hide

```
# Mapping high spots in clusters 1, 14 and the rest of the section.
mbp_colours <- c(
  "Cluster 1" = "#D55E00",
  "Cluster 14" = "#0072B2",
  "Other MBP-high" = "#756BB1",
  "Below cutoff" = "#D9D9D9"
)

mbp_spots$mbp_region <- factor(
  case_when(
    !mbp_spots$mbp_high ~ "Below cutoff",
    mbp_spots$cluster %in% focus ~ paste("Cluster", mbp_spots$cluster),
    TRUE ~ "Other MBP-high"
  ),
  levels = names(mbp_colours)
)

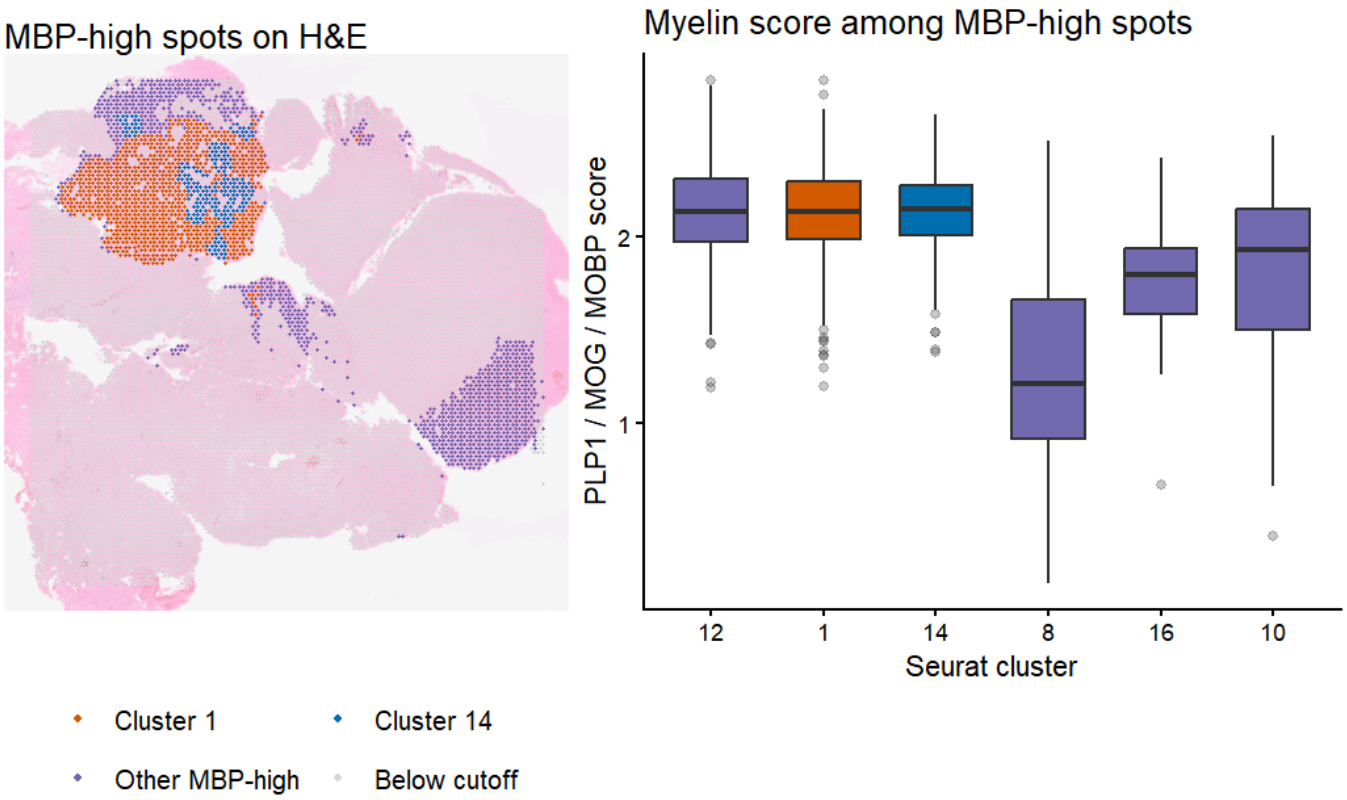
gbm_mbp_map <- AddMetaData(gbm_myelin, metadata = mbp_spots["mbp_region"])

p_mbp_map <- SpatialDimPlot(
  gbm_mbp_map,
  group.by = "mbp_region",
  images = Images(gbm_mbp_map)[1],
  image.scale = "lowres",
  cols = mbp_colours,
  pt.size.factor = 2.5,
  image.alpha = 0.55,
  combine = FALSE
)[[1]] +
  labs(title = "MBP-high spots on H&E", fill = NULL) +
  guides(fill = guide_legend(nrow = 2, byrow = TRUE)) +
  theme(
    legend.position = "bottom",
    legend.text = element_text(size = 10),
    legend.margin = margin(0, 0, 0, 0)
  )

# Comparing the score in clusters with enough high spots to show a distribution.
plot_clusters <- mbp_summary %>%
  filter(high_spots >= 20 | cluster %in% focus) %>%
  pull(cluster)

p_mbp_compare <- mbp_spots %>%
  filter(mbp_high, cluster %in% plot_clusters) %>%
  mutate(cluster = factor(cluster, levels = plot_clusters)) %>%
  ggplot(aes(cluster, MyelinCheck1, fill = mbp_region)) +
  geom_boxplot(width = 0.65, outlier.alpha = 0.25) +
  scale_fill_manual(values = mbp_colours, guide = "none") +
  labs(
    title = "Myelin score among MBP-high spots",
    x = "Seurat cluster",
    y = "PLP1 / MOG / MOBP score"
  ) +
  theme_classic()
```

```
p_mbp_map + p_mbp_compare
```



From the looks of it, cluster 12 demonstrates particularly similar Myelin scores among MBP-high spots and interestingly, at a fairly further region than cluster 1 and 14.

Posteriorly I will investigate what is the major difference between these two areas that drive the correlation score between them to be

0.45725566 (g1 vs g12) 0.43404108 (g12 vs g14)

Particularly when compared to the similarity if g1 vs g14

0.94376386 (g1 vs g14)

Hide

```
library(dplyr)

clusters <- c("1", "14", "12")
myelin_genes <- c("MBP", "PLP1", "MOG", "MOBP")

sct_residuals <- SeuratObject::LayerData(
  gbm_myelin, assay = "SCT", layer = "scale.data"
)

cluster_by_spot <- setNames(
  as.character(gbm_myelin$seurat_clusters),
  colnames(gbm_myelin)
)

spots_by_cluster <- split(
  colnames(sct_residuals),
  factor(cluster_by_spot[colnames(sct_residuals)], levels = clusters)
)

mbp_high_spots <- rownames(mbp_spots)[which(mbp_spots$mbp_high)]

data.frame(
  cluster = clusters,
  all_spots = lengths(spots_by_cluster),
  MBP_high_spots = lengths(lapply(spots_by_cluster, intersect, y = mbp_high_spots))
)
```

	cluster <chr>	all_spots <int>	MBP_high_spots <int>
1	1	969	891
14	14	272	236
12	12	478	456

3 rows

Hide

```
mean_profiles <- function(barcodes) {
  profiles <- vapply(
    spots_by_cluster,
    function(spots) {
      selected <- intersect(spots, barcodes)
      Matrix::rowMeans(sct_residuals[, selected, drop = FALSE])
    },
    numeric(nrow(sct_residuals))
  )
  rownames(profiles) <- rownames(sct_residuals)
  profiles
}

all_profiles <- mean_profiles(colnames(sct_residuals))
high_profiles <- mean_profiles(mbp_high_spots)
held_out <- !rownames(all_profiles) %in% myelin_genes

pairwise_rho <- function(profiles) {
  rho <- cor(profiles, method = "spearman", use = "pairwise.complete.obs")
  c(
    "12 vs 1" = rho["12", "1"],
    "12 vs 14" = rho["12", "14"],
    "1 vs 14" = rho["1", "14"]
  )
}

round(rbind(
  "All spots" = pairwise_rho(all_profiles),
  "All spots excluding myelin markers" = pairwise_rho(all_profiles[held_out, ]),
  "MBP-high spots" = pairwise_rho(high_profiles),
  "MBP-high spots excluding myelin markers" = pairwise_rho(high_profiles[held_out, ])
), 3)
```

```
                                    12 vs 1 12 vs 14 1 vs 14
All spots                             0.457    0.434   0.944
All spots excluding myelin markers    0.455    0.432   0.944
MBP-high spots                        0.493    0.468   0.939
MBP-high spots excluding myelin markers   0.491    0.465   0.939
```

Hide

```
# Find shared positive residuals and genes ranked differently in cluster 12.
gene_profiles <- tibble::tibble(
  gene = rownames(all_profiles),
  C1 = all_profiles[, "1"],
  C14 = all_profiles[, "14"],
  C12 = all_profiles[, "12"],
  shared_residual = pmin(C1, C14, C12),
  rank_shift_vs_1 = rank(C12) - rank(C1),
  rank_shift_vs_14 = rank(C12) - rank(C14)
)

shared_genes <- gene_profiles %>%
  filter(shared_residual > 0) %>%
  slice_max(shared_residual, n = 8, with_ties = FALSE) %>%
  select(gene, C1, C14, C12)

shifted_12 <- gene_profiles %>%
  filter(rank_shift_vs_1 * rank_shift_vs_14 > 0) %>%
  mutate(rank_shift = pmin(abs(rank_shift_vs_1), abs(rank_shift_vs_14))) %>%
  slice_max(rank_shift, n = 10, with_ties = FALSE) %>%
  select(gene, C1, C14, C12, rank_shift_vs_1, rank_shift_vs_14)

shared_genes
```

gene <chr>	C1 <dbl>	C14 <dbl>	C12 <dbl>
PLP1	9.820366	9.302750	9.036538
MBP	6.455420	5.986841	8.418800
MOBP	5.149794	5.004131	6.675297
CARNS1	4.121137	3.737067	3.880890
CNP	3.995665	3.699561	4.792416
TMEM144	3.704167	3.476139	4.027070
CLDND1	3.803625	3.888989	3.473692
ERMN	3.790241	3.470763	4.530164

8 rows

Hide

```
shifted_12
```

gene <chr>	C1 <dbl>	C14 <dbl>	C12 <dbl>	rank_shift_vs_1 <dbl>	rank_shift_vs_14 <dbl>
STAB1	3.775834	3.855923	-1.0602476	-2950	-2949
CD163	2.643443	3.121113	-1.6678239	-2948	-2961
CHI3L2	3.070306	2.854283	-1.1255423	-2946	-2930

gene <chr>	C1 <dbl>	C14 <dbl>	C12 <dbl>	rank_shift_vs_1 <dbl>	rank_shift_vs_14 <dbl>
PLTP	2.640767	2.975190	-1.1105233	-2922	-2931
KIF5A	-1.556623	-1.533512	1.8019306	2914	2915
C1QA	3.557624	3.545940	-0.8151792	-2904	-2903
CD14	2.513616	3.094833	-0.9716438	-2901	-2920
VSIG4	2.884711	3.196468	-0.8308524	-2891	-2895
MS4A6A	3.669975	3.897608	-0.7766367	-2890	-2893
C3	2.424955	2.438309	-0.8699562	-2878	-2875

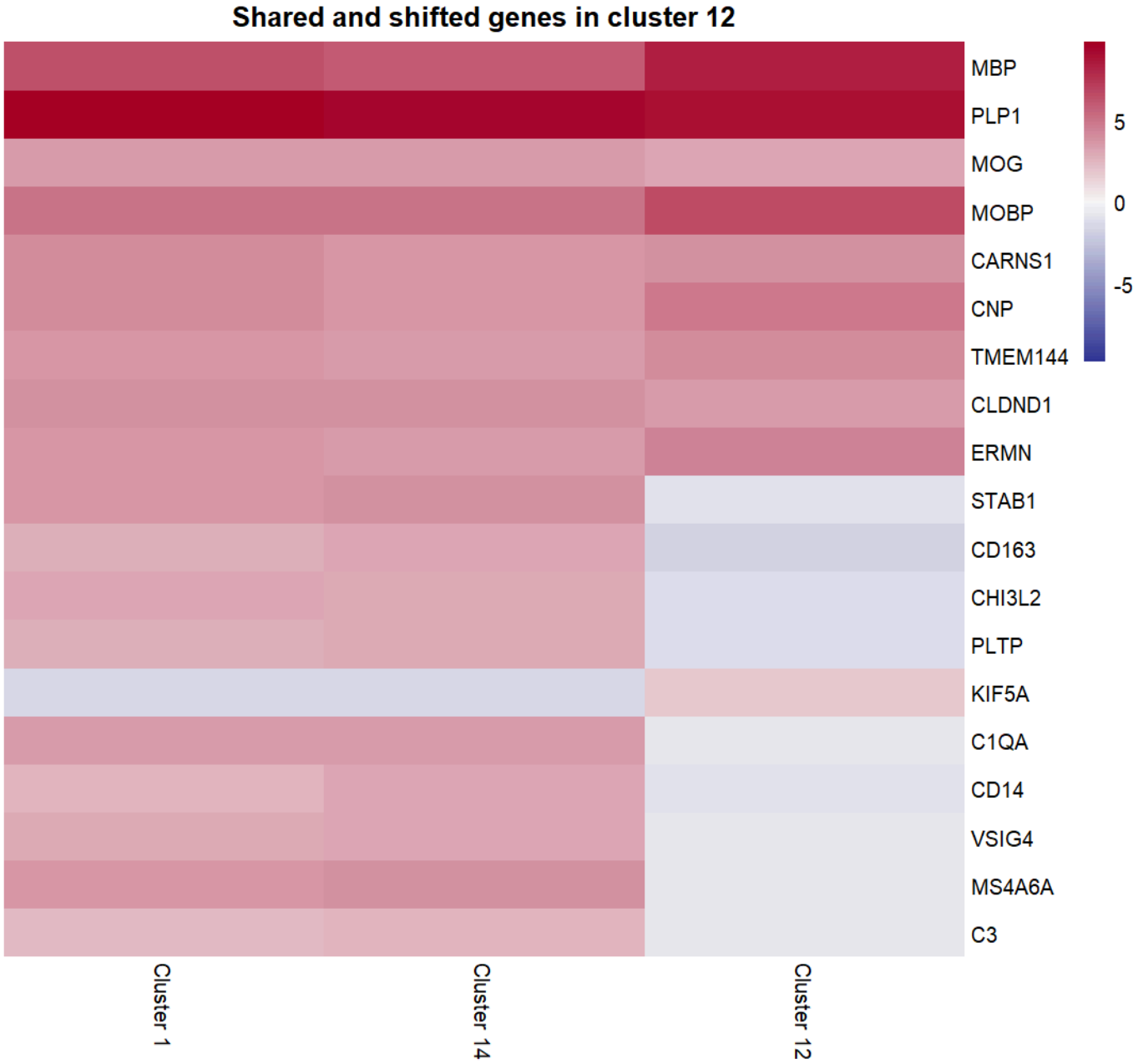
1-10 of 10 rows

Hide

```
genes_to_plot <- unique(c(
  intersect(myelin_genes, rownames(all_profiles)),
  shared_genes$gene,
  shifted_12$gene
))

heatmap_values <- all_profiles[genes_to_plot, , drop = FALSE]
colour_limit <- max(abs(heatmap_values))

pheatmap::pheatmap(
  heatmap_values,
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  labels_col = paste("Cluster", clusters),
  color = colorRampPalette(c("#313695", "#F7F7F7", "#A50026"))(101),
  breaks = seq(-colour_limit, colour_limit, length.out = 102),
  border_color = NA,
  main = "Shared and shifted genes in cluster 12"
)
```



Conclusions between cluster 1, 12 and 14

Cluster 1 and 14 present very similar genetic composition particularly within their predominant set of Myelin score.

Even though cluster 12 presents a fairly similar Myelin score within the MBP predominant spots of this tissue, this cluster demonstrates a shift towards lower genetic expression of genes that present higher SCT transformed RNA expression within cluster 1 and 14.

Interestingly a majority of cells within cluster 12 are actively expression KIF5A a marker that has previously been associated to correlate negatively with tumour grade, with lower levels of expression being highly present in GBM tumour cells (Joyce & Lindsay, 2024).

This could be a good explanation on why these clusters present such high similarity but also, somewhat dissimilar, with a high chance of a GBM tumour slowly forming within cluster 1 and 14, while cluster 12 cells demonstrate to be closely within healthy states.

Another major factor that may indicate this is the highly expression of CD14 and CD163 cells within cluster 1 and 14, indicating direct activity of immune system in this region and possibly part of a forming GBM.

Final Conclusions

Clusters 5 and 9 driving cluster 7 hypoxia and newly formed GBM.

With Clusters 5 and 9 being directly associated to fibrotic areas rich in collagen, this region may be directly correlated with a GBM formation in late necrotic stages, already forming collagen-rich repairing tissue.

Cluster 7 demonstrates particularly high expression of Hypoxic and Metastic genes, revealing a new area of metastasis forming within the region.

Clusters 1 and 14 Vs 12 (MBP)

Cluster 1 and 14 may represent a new cancer formation within the myelin rich areas of the brain, and even though distant from the main region of this tumour, this may be happening due to the metastatic nature of GBM.

Cluster 12 on the other hand, demonstrates particularly healthy levels when compared to the other two myelin rich clusters, mediated by lower levels of immune cell associated RNA expression, coupled with highly expressed KIF5A marker, which had a negative correlation to tumour grade.

References

Joyce, L.J. and Lindsay, A.J. (2024) ‘A systematic computational analysis of the endosomal recycling pathway in glioblastoma’, Biochemistry and Biophysics Reports, 38, p. 101700. doi:10.1016/j.bbrep.2024.101700 (doi:10.1016/j.bbrep.2024.101700).